[GETNET]INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

Projeto do PRM contempla 2 RACs, um RAC Ativo no DataCenter Sul e outro RAC no DataCenter Norte. Entre estes dois RACs haverá um Data Guard.

RAC Sul

- GNCASSPL08359
- GNCASSPL08360

RAC Norte

- GNCASNPL08358
- GNCASNPL08361

Este documento registra a instalação do RAC no Norte.

Entrega dos servidores

Servidores foram entregues pelo João Hoffman, tratam-se de máquinas físicas, sendo assim, revisamos diversos paramêtros de memoria, users e demais requisitos Oracle. Ao desenrolar a investigação, notamos que havia um StanAlone instalado em cada uma das máquinas.

Parâmetros de memória:

```
[root@gncasnpl08361 ~]# cat /etc/sysctl.conf
# System default settings live in /usr/lib/sysctl.d/00-system.conf.
# To override those settings, enter new settings here, or in an /etc/sysctl.d/<name>.conf file
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
# Controls IP packet forwarding
net.ipv4.ip_forward = 0

# Controls source route verification
net.ipv4.conf.default.rp_filter = 1

# Do not accept source routing
net.ipv4.conf.default.accept_source_route = 0

# Controls the System Request debugging functionality of the kernel
kernel.sysrq = 0

# Controls whether core dumps will append the PID to the core filename.
# Useful for debugging multi-threaded applications.
kernel.core_uses_pid = 1

# Controls the use of TCP syncookies
net.ipv4.tcp_syncookies = 1

# Disable netfilter on bridges.
net.bridge.bridge-nf-call-ip6tables = 0
net.bridge.bridge-nf-call-iptables = 0
net.bridge.bridge-nf-call-arptables = 0

# Controls the default maxmimum size of a mesage queue
kernel.msgmnb = 65536

# Controls the maximum size of a message, in bytes
kernel.msgmax = 65536
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmall = 4294967296
kernel.shmmni = 4096
kernel.sem = 250 32000 100 256
net.ipv4.ip_local_port_range = 9000 65500
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048586
kernel.shmmax = 270377232384
```

Todos os pacotes necessários, executar nos dois nós. Documentei apenas 1 deles.

```
[root@gncasnpl08358 ~]# dnf install bc binutils elfutils-libelf elfutils-libelf-devel fontconfig-devel glibc glibc-devel ksh libaio libaio-devel libgcc libnsl librdmacm-devel libstdc++ libstdc++-devel libX11 libXau libxcb libXi libXrender libXrender-devel libXtst make net-tools nfs-utils python3 python3-configshell python3-rtslib python3-six smartmontools sysstat targetcli unzip xorg-x11-apps chrony tmux xorg-x11-server-utils glibc.i686 glibc-devel.i686 libstdc++-devel.i686 libaio-devel.i686 libXext.i686 libXtst.i686 zlib-devel.i686 -y

Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs)            54 kB/s | 2.8 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs)    43 kB/s | 2.4 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs)               47 kB/s | 2.4 kB     00:00
Zabbix 5.0 frontend                                                 26 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 I386                                                 41 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 x86_64                                               44 kB/s | 2.1 kB     00:00
Zabbix - Rhel7                                                      28 kB/s | 2.1 kB     00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs)                44 kB/s | 2.1 kB     00:00
Zabbix 6.0                                                          44 kB/s | 2.1 kB     00:00
Package bc-1.07.1-5.el8.x86_64 is already installed.
Package binutils-2.30-119.el8.x86_64 is already installed.
Package elfutils-libelf-0.188-3.el8.x86_64 is already installed.
Package elfutils-libelf-devel-0.188-3.el8.x86_64 is already installed.
Package fontconfig-devel-2.13.1-4.el8.x86_64 is already installed.
Package glibc-2.28-225.el8.x86_64 is already installed.
Package glibc-devel-2.28-225.el8.x86_64 is already installed.
Package ksh-20120801-257.el8.x86_64 is already installed.
Package libaio-0.3.112-1.el8.x86_64 is already installed.
Package libaio-devel-0.3.112-1.el8.x86_64 is already installed.
Package libgcc-8.5.0-18.el8.x86_64 is already installed.
Package libnsl-2.28-225.el8.x86_64 is already installed.
Package rdma-core-devel-44.0-2.el8.1.x86_64 is already installed.
Package libstdc++-8.5.0-18.el8.x86_64 is already installed.
Package libstdc++-devel-8.5.0-18.el8.x86_64 is already installed.
Package libX11-1.6.8-5.el8.x86_64 is already installed.
Package libXau-1.0.9-3.el8.x86_64 is already installed.
Package libxcb-1.13.1-1.el8.x86_64 is already installed.
Package libXi-1.7.10-1.el8.x86_64 is already installed.
Package libXrender-0.9.10-7.el8.x86_64 is already installed.
Package libXrender-devel-0.9.10-7.el8.x86_64 is already installed.
Package libXtst-1.2.3-7.el8.x86_64 is already installed.
Package make-1:4.2.1-11.el8.x86_64 is already installed.
Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Package nfs-utils-1:2.3.3-59.el8.x86_64 is already installed.
Package python36-3.6.8-38.module+el8.5.0+12207+5c5719bc.x86_64 is already installed.
Package python3-six-1.11.0-8.el8.noarch is already installed.
Package smartmontools-1:7.1-1.el8.x86_64 is already installed.
Package sysstat-11.7.3-9.el8.x86_64 is already installed.
Package unzip-6.0-46.el8.x86_64 is already installed.
No match for argument: xorg-x11-apps
Package chrony-4.2-1.el8.x86_64 is already installed.
Package xorg-x11-server-utils-7.7-27.el8.x86_64 is already installed.
Error: Unable to find a match: xorg-x11-apps

[root@gncasnpl08358 ~]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs)            51 kB/s | 2.8 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs)    48 kB/s | 2.4 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs)               50 kB/s | 2.4 kB     00:00
Zabbix 5.0 frontend                                                 42 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 I386                                                 43 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 x86_64                                               42 kB/s | 2.1 kB     00:00
Zabbix - Rhel7                                                      44 kB/s | 2.1 kB     00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs)                42 kB/s | 2.1 kB     00:00
Zabbix 6.0                                                          37 kB/s | 2.1 kB     00:00
Package xorg-x11-xauth-1:1.0.9-12.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!

[root@gncasnpl08358 ~]# yum install xterm
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs)            43 kB/s | 2.8 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs)    47 kB/s | 2.4 kB     00:00
```

```
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs)          47 kB/s | 2.4 kB     00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs)                     50 kB/s | 2.4 kB     00:00
Zabbix 5.0 frontend                                                       42 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 I386                                                       44 kB/s | 2.1 kB     00:00
Zabbix - Rhel6 x86_64                                                     41 kB/s | 2.1 kB     00:00
Zabbix - Rhel7                                                            44 kB/s | 2.1 kB     00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs)                      43 kB/s | 2.1 kB     00:00
Zabbix 6.0                                                                44 kB/s | 2.1 kB     00:00
Dependencies resolved.
==================================================================================================
 Package                  Architecture   Version         Repository                        Size
==================================================================================================
Installing:
 xterm                    x86_64         331-1.el8_3.2   rhel-8-for-x86_64-appstream-rpms   529 k
Installing dependencies:
 libXaw                   x86_64         1.0.13-10.el8   rhel-8-for-x86_64-appstream-rpms   194 k
 xterm-resize             x86_64         331-1.el8_3.2   rhel-8-for-x86_64-appstream-rpms    38 k
Installing weak dependencies:
 xorg-x11-fonts-misc      noarch         7.5-19.el8      rhel-8-for-x86_64-appstream-rpms   5.8 M

Transaction Summary
==================================================================================================
Install  4 Packages

Total download size: 6.5 M
Installed size: 8.8 M
Is this ok [y/N]: y
Downloading Packages:
(1/4): xterm-331-1.el8_3.2.x86_64.rpm                               5.3 MB/s | 529 kB     00:00
(2/4): xterm-resize-331-1.el8_3.2.x86_64.rpm                        1.2 MB/s |  38 kB     00:00
(3/4): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm                     26 MB/s | 5.8 MB     00:00
(4/4): libXaw-1.0.13-10.el8.x86_64.rpm                               32 kB/s | 194 kB     00:06
--------------------------------------------------------------------------------------------------
Total                                                               1.1 MB/s | 6.5 MB     00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Installing       : xterm-resize-331-1.el8_3.2.x86_64                                      1/4
  Installing       : libXaw-1.0.13-10.el8.x86_64                                            2/4
  Installing       : xorg-x11-fonts-misc-7.5-19.el8.noarch                                  3/4
  Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch                                  3/4
  Installing       : xterm-331-1.el8_3.2.x86_64                                             4/4
  Running scriptlet: xterm-331-1.el8_3.2.x86_64                                             4/4
  Verifying        : xorg-x11-fonts-misc-7.5-19.el8.noarch                                  1/4
  Verifying        : libXaw-1.0.13-10.el8.x86_64                                            2/4
  Verifying        : xterm-331-1.el8_3.2.x86_64                                             3/4
  Verifying        : xterm-resize-331-1.el8_3.2.x86_64                                      4/4
Installed products updated.

Installed:
  libXaw-1.0.13-10.el8.x86_64     xorg-x11-fonts-misc-7.5-19.el8.noarch     xterm-331-1.el8_3.2.x86_64     xterm-resize-331-1.el8_3.2.x86_64

Complete!
[root@gncasnpl08358 ~]#
```

Users oracle e grid já estavam configurados também, nos dois nos:

```
[root@gncasnpl08358 ~]# id oracle
uid=54340(oracle) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]# id grid
uid=54341(grid) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]#
```

Disable firewall, nos dois nos:

```
[root@gncasnpl08358 ~]# systemctl stop firewalld
[root@gncasnpl08358 ~]#
[root@gncasnpl08358 ~]# systemctl disable firewalld
Removed /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
```

Criação de profiles. Lembre que o nó 1 é PRMPRD1 e ASM1 e nó 2 PRMPRD2 e ASM2:

```
vi /home/oracle/db_profile

export ORACLE_BASE=/u01/app/oracle_19c
export ORACLE_HOME=/u01/app/oracle/product/19.19.0/dbhome_1
export ORACLE_SID=PRMCTG1
export OH=$ORACLE_HOME
export JAVA_HOME=/usr/bin/java
export GGS_HOME=/ggs
export PATH=.:/usr/bin/java/bin:/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/oracle/bin:$ORACLE_HOME/bin:$ORACLE_HOME/OPatc
h:$PATH
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib:$ORACLE_HOME/network/jlib
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$ORACLE_HOME/oracm/lib:/lib:/usr/lib:/usr/local/lib
export LIBPATH=/ggs:$ORACLE_HOME/lib:/usr/lib:$LIBPATH

chmod 775 /home/oracle/db_profile
chown oracle:oinstall /home/oracle/db_profile
vi /home/grid/grid_profile

export ORACLE_BASE=/u01/app/grid_19c
export ORACLE_HOME=/u01/app/19.19.0/grid
export ORACLE_SID=+ASM1;
export OH=$ORACLE_HOME
export PATH=$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib

chmod 775 /home/grid/grid_profile
chown grid:oinstall /home/grid/grid_profile
```

Download e unzip dos pacotes, pode baixar nos 2 nós, mas os unzip da home do grid e oracle, somente no nó 01:

```
su - grid
. /home/grid/grid_profile
cd /u02/oracle/patches

wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35037840_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35050341_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35004974_190000_Linux-x86-64.zip
```

oracleasm

João já entregou as máquinas com tudo instalado, sendo assim, vamos ter que desinstalar tudo e seguir exatamente como fizemos nos RACs de HML.

```
[root@gncasnpl08358 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13

[root@gncasnpl08361 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```

Shutdown dos serviços e deinstall

```
[grid@gncasnpl08358 app]$ . /home/grid/grid_profile
[grid@gncasnpl08358 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08358' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08358' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358' has completed
CRS-4133: Oracle High Availability Services has been stopped.

[grid@gncasnpl08361 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08361' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08361' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361' has completed
CRS-4133: Oracle High Availability Services has been stopped.
```

```
[root@gncasnpl08358 ~]# . /home/grid/grid_profile
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]#

[root@gncasnpl08361 ~]# . /home/grid/grid_profile
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
```

```
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]#

[root@gncasnpl08358 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 ~]# oracleasm listdisks

[root@gncasnpl08361 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08361 ~]# oracleasm listdisks
```

Remoção dos binários:

```
[root@gncasnpl08358 ~]# cd /tmp
[root@gncasnpl08358 tmp]# rm -rf *
[root@gncasnpl08358 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08358 tmp]# cd /etc/oracle
[root@gncasnpl08358 oracle]# rm -rf *
[root@gncasnpl08358 oracle]# rm -rf /etc/oratab
[root@gncasnpl08358 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08358 oracle]#  /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08358 oracle]#  rmrole oracle_devmgmt
```

```
bash: rmrole: command not found...
[root@gncasnpl08358 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08358 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08358 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08358 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08358 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]#  cd /u01/
[root@gncasnpl08358 u01]# ls -la
total 28
drwxr-xr-x   4 oracle oinstall  4096 Jun 19 09:20 .
dr-xr-xr-x. 22 root   root      4096 Jul 18 16:47 ..
drwxrwx---   8 oracle oinstall  4096 Jul 20 11:40 app
drwx------   2 root   root     16384 Jun 16 17:49 lost+found
[root@gncasnpl08358 u01]# rm -rf app
[root@gncasnpl08358 u01]#

[root@gncasnpl08361 ~]# cd /tmp
[root@gncasnpl08361 tmp]# rm -rf *
[root@gncasnpl08361 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08361 tmp]# cd /etc/oracle
[root@gncasnpl08361 oracle]# rm -rf *
[root@gncasnpl08361 oracle]# rm -rf /etc/oratab
[root@gncasnpl08361 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08361 oracle]#  /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08361 oracle]#  rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08361 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08361 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08361 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08361 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08361 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]#  cd /u01/
[root@gncasnpl08361 u01]# ls -la
total 28
drwxr-xr-x   4 oracle oinstall  4096 Jun 19 14:45 .
dr-xr-xr-x. 22 root   root      4096 Jun 16 17:38 ..
drwxrwx---   8 oracle oinstall  4096 Jul 20 11:41 app
drwx------   2 root   root     16384 Jun 16 17:50 lost+found
[root@gncasnpl08361 u01]# rm -rf app
[root@gncasnpl08361 u01]#
```

Criação dos novos diretórios:

```
mkdir -p /u01/app/oracle/product/19.19.0/dbhome_1
mkdir -p /u01/app/19.19.0/grid
chown -R grid:oinstall /u01/app/19.19.0/
chown -R oracle:oinstall /u01/app/oracle/product/19.19.0
mkdir -p /u01/app/grid_19c
mkdir -p /u01/app/oracle_19c
chown -R grid:oinstall /u01/app/grid_19c/
chown -R oracle:oinstall /u01/app/oracle_19c/
chmod 775 /u01/app/oracle_19c/
chmod 775 /u01/app/grid_19c/
```

Formatação dos discos (que foram apagados previamente do ASM):

```
[root@gncasnpl08358 u01]# fdisk -l
Disk /dev/sda: 446.6 GiB, 479559942144 bytes, 936640512 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disklabel type: gpt
Disk identifier: EFD0B073-FC44-4EF7-B31F-0E3BA6ACFD10

Device       Start       End   Sectors   Size Type
/dev/sda1     2048    411647    409600   200M EFI System
/dev/sda2   411648   4605951   4194304     2G Microsoft basic data
/dev/sda3  4605952   6703103   2097152     1G Linux filesystem
/dev/sda4  6703104 936638463 929935360 443.4G Linux LVM


Disk /dev/sdb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdb1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdc1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdd: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdd1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sde: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a

Device     Boot Start       End   Sectors  Size Id Type
```

```
/dev/sde1        2048 629145599 629143552  300G 83 Linux


Disk /dev/sdf: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdf1        2048 629145599 629143552  300G 83 Linux


Disk /dev/sdg: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdh: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26

Device     Boot Start      End  Sectors Size Id Type
/dev/sdh1        2048 83888639 83886592  40G 83 Linux


Disk /dev/sdi: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040

Device     Boot Start      End  Sectors Size Id Type
/dev/sdi1        2048 83888639 83886592  40G 83 Linux


Disk /dev/sdj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdj1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdk1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
```

```
Device     Boot Start        End    Sectors  Size Id Type
/dev/sdl1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdm: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdm1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdn: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdn1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdo: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdo1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdp: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdp1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdq1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdr1        2048 2147485439 2147483392 1024G 83 Linux
```

```
Disk /dev/sds: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581

Device     Boot Start        End    Sectors  Size Id Type
/dev/sds1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdt: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdt1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdu1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdv1        2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdw: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdw1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdx: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdx1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdy: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdy1        2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdz: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdz1        2048 629145599 629143552  300G 83 Linux


Disk /dev/sdaa: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdaa1       2048 629145599 629143552  300G 83 Linux


Disk /dev/sdab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdac: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26

Device     Boot Start      End  Sectors Size Id Type
/dev/sdac1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sdad: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040

Device     Boot Start      End  Sectors Size Id Type
/dev/sdad1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sdae: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdae1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdaf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
```

```
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdaf1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdag1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdah1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdai: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdai1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdaj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdaj1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdak: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdak1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdal: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
```

```
Device     Boot Start        End    Sectors  Size Id Type
/dev/sdal1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdam: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdam1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdan: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdan1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdao: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdao1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdap: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdap1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdaq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdaq1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdar: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdar1       2048 301992959 301990912  144G 8e Linux LVM
```

```
Disk /dev/sdas: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdas1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdat: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdat1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdau: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdau1       2048 629145599 629143552  300G 83 Linux


Disk /dev/sdav: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdav1       2048 629145599 629143552  300G 83 Linux


Disk /dev/sdaw: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdax: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26

Device     Boot Start      End  Sectors Size Id Type
/dev/sdax1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sday: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040

Device     Boot Start      End  Sectors Size Id Type
```

```
/dev/sday1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sdaz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdaz1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdba: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdba1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbb1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbc1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbd1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbe: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbe1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
```

```
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbf1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbg1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbh: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbh1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbi: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbi1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbj1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbk1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
```

```
Disk identifier: 0xcfb47f11

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbl1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbm: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdbm1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdbn: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdbn1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdbo: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdbo1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/sdbp: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdbp1       2048 629145599 629143552  300G 83 Linux


Disk /dev/sdbq: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdbq1       2048 629145599 629143552  300G 83 Linux


Disk /dev/sdbr: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdbs: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26

Device     Boot Start      End  Sectors Size Id Type
/dev/sdbs1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sdbt: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040

Device     Boot Start      End  Sectors Size Id Type
/dev/sdbt1       2048 83888639 83886592  40G 83 Linux


Disk /dev/sdbu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbu1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbv1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbw1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbx1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdby: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
```

```
Device     Boot Start        End    Sectors  Size Id Type
/dev/sdby1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdbz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdbz1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdca: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdca1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdcb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdcb1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdcc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdcc1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdcd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdcd1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdce: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdce1       2048 2147485439 2147483392 1024G 83 Linux
```

```
Disk /dev/sdcf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdcf1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/sdcg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11

Device     Boot Start        End    Sectors  Size Id Type
/dev/sdcg1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/rhel-root: 70 GiB, 75161927680 bytes, 146800640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes


Disk /dev/mapper/rhel-swap: 4 GiB, 4294967296 bytes, 8388608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes


Disk /dev/mapper/rhel-home: 369.4 GiB, 396667912192 bytes, 774742016 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes


Disk /dev/mapper/mpatha: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1

Device              Boot Start       End   Sectors  Size Id Type
/dev/mapper/mpatha1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/mapper/mpathb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a

Device              Boot Start       End   Sectors  Size Id Type
/dev/mapper/mpathb1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/mapper/mpathq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd

Device              Boot Start        End    Sectors  Size Id Type
```

```
/dev/mapper/mpathq1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathr1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/vgu01-u01: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/vgu02-u02: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/mpaths: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpaths1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpatht: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpatht1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathu1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathv1       2048 2147485439 2147483392 1024G 83 Linux
```

```
Disk /dev/mapper/mpathw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathw1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathx1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathy: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathy1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathz1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c

Device              Boot Start       End   Sectors  Size Id Type
/dev/mapper/mpathc1       2048 301992959 301990912  144G 8e Linux LVM


Disk /dev/mapper/mpathaa: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11

Device               Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathaa1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
Disk identifier: 0xf767a22a

Device              Boot Start       End   Sectors  Size Id Type
/dev/mapper/mpathab1       2048 629145599 629143552  300G 83 Linux


Disk /dev/mapper/vg_ggs-ggs: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/mpathac: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd

Device              Boot Start       End   Sectors  Size Id Type
/dev/mapper/mpathac1       2048 629145599 629143552  300G 83 Linux


Disk /dev/mapper/mpathad: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/mpathae: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26

Device              Boot Start      End  Sectors Size Id Type
/dev/mapper/mpathae1       2048 83888639 83886592  40G 83 Linux


Disk /dev/mapper/mpathaf: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040

Device              Boot Start      End  Sectors Size Id Type
/dev/mapper/mpathaf1       2048 83888639 83886592  40G 83 Linux


Disk /dev/mapper/mpathag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac

Device              Boot Start        End    Sectors  Size Id Type
/dev/mapper/mpathag1       2048 2147485439 2147483392 1024G 83 Linux


Disk /dev/mapper/mpathah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
```

```
Device             Boot Start       End    Sectors  Size Id Type
/dev/mapper/mpathah1       2048 2147485439 2147483392 1024G 83 Linux
[root@gncasnpl08358 u01]#
```

Output do multipath

```
[root@gncasnpl08358 u01]# multipath -ll
mpatha (360000970000597600012533030304431) dm-3 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:0  sdb  8:16   active ready running
  |- 11:0:1:0  sdw  65:96  active ready running
  |- 13:0:0:0  sdar 66:176 active ready running
  `- 13:0:1:0  sdbm 68:0   active ready running
mpathaa (360000970000597600012533030313542) dm-20 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:20 sdv  65:80  active ready running
  |- 11:0:1:20 sdaq 66:160 active ready running
  |- 13:0:0:20 sdbl 67:240 active ready running
  `- 13:0:1:20 sdcg 69:64  active ready running
mpathab (360000970000597600012533030303435) dm-22 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:3  sde  8:64   active ready running
  |- 11:0:1:3  sdz  65:144 active ready running
  |- 13:0:0:3  sdau 66:224 active ready running
  `- 13:0:1:3  sdbp 68:48  active ready running
mpathac (360000970000597600012533030303436) dm-24 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:4  sdf  8:80   active ready running
  |- 11:0:1:4  sdaa 65:160 active ready running
  |- 13:0:0:4  sdav 66:240 active ready running
  `- 13:0:1:4  sdbq 68:64  active ready running
mpathad (360000970000597600012533030313443) dm-25 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:5  sdg  8:96   active ready running
  |- 11:0:1:5  sdab 65:176 active ready running
  |- 13:0:0:5  sdaw 67:0   active ready running
  `- 13:0:1:5  sdbr 68:80  active ready running
mpathae (360000970000597600012533030313444) dm-26 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:6  sdh  8:112  active ready running
  |- 11:0:1:6  sdac 65:192 active ready running
  |- 13:0:0:6  sdax 67:16  active ready running
  `- 13:0:1:6  sdbs 68:96  active ready running
mpathaf (360000970000597600012533030313445) dm-27 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:7  sdi  8:128  active ready running
  |- 11:0:1:7  sdad 65:208 active ready running
  |- 13:0:0:7  sday 67:32  active ready running
  `- 13:0:1:7  sdbt 68:112 active ready running
mpathag (360000970000597600012533030313446) dm-28 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:8  sdj  8:144  active ready running
  |- 11:0:1:8  sdae 65:224 active ready running
  |- 13:0:0:8  sdaz 67:48  active ready running
  `- 13:0:1:8  sdbu 68:128 active ready running
mpathah (360000970000597600012533030313530) dm-29 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:9  sdk  8:160  active ready running
  |- 11:0:1:9  sdaf 65:240 active ready running
  |- 13:0:0:9  sdba 67:64  active ready running
  `- 13:0:1:9  sdbv 68:144 active ready running
```

```
mpathb (360000970000597600012533030304432) dm-4 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:1  sdc  8:32   active ready running
  |- 11:0:1:1  sdx  65:112 active ready running
  |- 13:0:0:1  sdas 66:192 active ready running
  `- 13:0:1:1  sdbn 68:16  active ready running
mpathc (360000970000597600012533030304433) dm-19 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:2  sdd  8:48   active ready running
  |- 11:0:1:2  sdy  65:128 active ready running
  |- 13:0:0:2  sdat 66:208 active ready running
  `- 13:0:1:2  sdbo 68:32  active ready running
mpathq (360000970000597600012533030313531) dm-6 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:10 sdl  8:176  active ready running
  |- 11:0:1:10 sdag 66:0   active ready running
  |- 13:0:0:10 sdbb 67:80  active ready running
  `- 13:0:1:10 sdbw 68:160 active ready running
mpathr (360000970000597600012533030313532) dm-8 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:11 sdm  8:192  active ready running
  |- 11:0:1:11 sdah 66:16  active ready running
  |- 13:0:0:11 sdbc 67:96  active ready running
  `- 13:0:1:11 sdbx 68:176 active ready running
mpaths (360000970000597600012533030313533) dm-11 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:12 sdn  8:208  active ready running
  |- 11:0:1:12 sdai 66:32  active ready running
  |- 13:0:0:12 sdbd 67:112 active ready running
  `- 13:0:1:12 sdby 68:192 active ready running
mpatht (360000970000597600012533030313534) dm-12 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:13 sdo  8:224  active ready running
  |- 11:0:1:13 sdaj 66:48  active ready running
  |- 13:0:0:13 sdbe 67:128 active ready running
  `- 13:0:1:13 sdbz 68:208 active ready running
mpathu (360000970000597600012533030313535) dm-13 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:14 sdp  8:240  active ready running
  |- 11:0:1:14 sdak 66:64  active ready running
  |- 13:0:0:14 sdbf 67:144 active ready running
  `- 13:0:1:14 sdca 68:224 active ready running
mpathv (360000970000597600012533030313536) dm-14 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:15 sdq  65:0   active ready running
  |- 11:0:1:15 sdal 66:80  active ready running
  |- 13:0:0:15 sdbg 67:160 active ready running
  `- 13:0:1:15 sdcb 68:240 active ready running
mpathw (360000970000597600012533030313537) dm-15 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:16 sdr  65:16  active ready running
  |- 11:0:1:16 sdam 66:96  active ready running
  |- 13:0:0:16 sdbh 67:176 active ready running
  `- 13:0:1:16 sdcc 69:0   active ready running
mpathx (360000970000597600012533030313538) dm-16 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:17 sds  65:32  active ready running
  |- 11:0:1:17 sdan 66:112 active ready running
  |- 13:0:0:17 sdbi 67:192 active ready running
  `- 13:0:1:17 sdcd 69:16  active ready running
```

```
mpathy (360000970000597600012533030313539) dm-17 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:18 sdt  65:48  active ready running
  |- 11:0:1:18 sdao 66:128 active ready running
  |- 13:0:0:18 sdbj 67:208 active ready running
  `- 13:0:1:18 sdce 69:32  active ready running
mpathz (360000970000597600012533030313541) dm-18 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
  |- 11:0:0:19 sdu  65:64  active ready running
  |- 11:0:1:19 sdap 66:144 active ready running
  |- 13:0:0:19 sdbk 67:224 active ready running
  `- 13:0:1:19 sdcf 69:48  active ready running
[root@gncasnpl08358 u01]#
```

Apagamos apenas os discos que foram utilizados na instalação standalone e reapresentamos ao ASM:

```
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathae1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00409259 s, 12.5 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaf1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00649853 s, 7.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathab1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00404717 s, 12.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathac1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00418631 s, 12.2 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathad1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.000105445 s, 486 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathq1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00391171 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathr1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00395635 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpaths1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00358853 s, 14.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpatht1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00451208 s, 11.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathu1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00439451 s, 11.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathv1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00373145 s, 13.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathw1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00607942 s, 8.4 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathx1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00389783 s, 13.1 MB/s
```

```
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathy1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00399422 s, 12.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathz1  bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00397336 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaa1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00370638 s, 13.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathag1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00364786 s, 14.0 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathah1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00387046 s, 13.2 MB/s
[root@gncasnpl08358 u01]#



[root@gncasnpl08358 u01]# oracleasm createdisk  ARCH_PRM84PRD01 /dev/mapper/mpathae1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  ARCH_PRM84PRD02 /dev/mapper/mpathaf1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_AUX_PRM84PRD01 /dev/mapper/mpathab1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_AUX_PRM84PRD02 /dev/mapper/mpathac1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD01 /dev/mapper/mpathq1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD02 /dev/mapper/mpathr1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD03 /dev/mapper/mpaths1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD04 /dev/mapper/mpatht1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD05 /dev/mapper/mpathu1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD06 /dev/mapper/mpathv1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD07 /dev/mapper/mpathw1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD08 /dev/mapper/mpathx1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD09 /dev/mapper/mpathy1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD10 /dev/mapper/mpathz1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD11 /dev/mapper/mpathaa1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD12 /dev/mapper/mpathag1
```

```
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk  DATA_PRM84PRD13 /dev/mapper/mpathah1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]#


[root@gncasnpl08358 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...

[root@gncasnpl08358 u01]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```

Agora temos discos de ASM zerados e apresentados ao nó 01.
Podemos seguir com as questões de interfaces de rede. Consolidação dos IPs:

External Image
External Image
Resumo da configuração dos IPs:

```
GNCASNPL08358
IP Público:  180.104.147.12
IP Branca:  22.248.70.14
IP Backup:  22.240.70.12
Interconnect:  172.27.147.11
IP VIP:  180.104.147.15

Scan1:  180.104.147.19
Scan2:  180.104.147.20
Scan3:  180.104.147.21

GNCASNPL08361
IP Público:  180.104.147.14
IP Branca:  22.248.70.16
IP Backup:  22.240.70.15
Interconnect:  172.27.147.12
IP VIP:  180.104.147.18
```

Resolução das chaves de SSH:
NOTA: o documento de RAC de HML está completo a parte do SSH.

```
passwd oracle
passwd grid
*** usar uma senha padrão

./sshUserSetup.sh -user grid -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
./sshUserSetup.sh -user oracle -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase

chmod 600 /home/grid/.ssh/config
chmod 600 /home/oracle/.ssh/config
```

Configuração do /etc/hosts:

```
### Install RAC Oracle
180.104.147.12    GNCASNPL08358      GNCASNPL08358.getnet.local
22.240.70.12      GNCASNPL08358.bkp  GNCASNPL08358.bkp.getnet.local
22.248.70.14      GNCASNPL08358.wt   GNCASNPL08358.wt.getnet.local
180.104.147.15    GNCASNPL08358-vip  GNCASNPL08358-vip.getnet.local

180.104.147.14    GNCASNPL08361      GNCASNPL08361.getnet.local
22.240.70.15      GNCASNPL08361.bkp  GNCASNPL08361.bkp.getnet.local
22.248.70.16      GNCASNPL08361.wt   GNCASNPL08361.wt.getnet.local
180.104.147.18    GNCASNPL08361-vip  GNCASNPL08361-vip.getnet.local
```

Download do patch que corrige o SSH:

```
su - grid
. grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_Linux-x86-64.zip
unzip p30159782_195000OCWRU_Linux-x86-64.zip
```

Install Grid Infrastructure

```
[grid@gncasnpl08358 ~]$ . grid_profile
[grid@gncasnpl08358 ~]$ cd $ORACLE_HOME
[grid@gncasnpl08358 grid]$ xauth add gncasnpl08358/unix:11  MIT-MAGIC-COOKIE-1  61ffaebd4b2496351672b043d28ab8fc
[grid@gncasnpl08358 grid]$ export DISPLAY=localhost:11.0
[grid@gncasnpl08358 grid]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 grid]$ ./gridSetup.sh -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /tmp/GridSetupActions2023-07-20_04-46-39PM/installerPatchActions_2023-07-20_04-46-39PM.log
Launching Oracle Grid Infrastructure Setup Wizard...

[FATAL] [INS-40426] Grid installation option has not been specified.
   ACTION: Specify the valid installation option.

[grid@gncasnpl08358 grid]$ ./gridSetup.sh
```

External Image
External Image
External Image
Adicionar o segundo nó do cluster:
Clique em “Add…”

External Image
External Image
External Image
Como trata-se de máquinas físicas, há mais interfaces de redes configuradas, como da idrac, por exemplo, lembre-se de utilizar a interface publica para o public e interconnect para ASM & Private:

External Image
NOTA: O erro abaixo está relacionado a interface do IDrac, conversei com o João HOffman, ele me pediu para ignorar:

External Image
External Image
External Image
External Image
NOTA: lembre-se de ajustar o discovery path:

External Image
Escolhemos apenas discos do +DATA:

External Image
External Image
External Image
External Image
External Image
External Image
External Image
External Image
Observações:
Node Connectivity é a respeito do Idrac, o swap eu combinei do João Hoffman ajustar posteriormente dado a emergencia do projeto e os demais são irrelevantes:

External Image
External Image
External Image

Execução de scripts:

```
[root@gncasnpl08358 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.

[root@gncasnpl08361 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.

[root@gncasnpl08358 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= grid
    ORACLE_HOME=  /u01/app/19.19.0/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.


Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
  /u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/rootcrs_gncasnpl08358_2023-07-25_05-58-37PM.log
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 17:58:46 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 17:58:47 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 17:58:56 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 17:58:59 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 17:59:07 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 17:59:14 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 17:59:14 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/07/25 17:59:34 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 17:59:38 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 17:59:42 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 17:59:45 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.

ASM has been created and started successfully.

[DBT-30001] Disk groups created successfully. Check /u01/app/grid_19c/cfgtoollogs/asmca/asmca-230725PM060015.log for details.

2023/07/25 18:01:21 CLSRSC-482: Running command: '/u01/app/19.19.0/grid/bin/ocrconfig -upgrade grid oinstall'
CRS-4256: Updating the profile
Successful addition of voting disk 25590c0c233d4f26bf21c16dcb0516a4.
Successfully replaced voting disk group with +DATA_PRM84PRD.
CRS-4256: Updating the profile
```

```
CRS-4266: Voting file(s) successfully replaced
##  STATE    File Universal Id                File Name Disk group
--  -----    -----------------                --------- ---------
 1. ONLINE   25590c0c233d4f26bf21c16dcb0516a4 (/dev/oracleasm/disks/DATA_PRM84PRD01) [DATA_PRM84PRD]
Located 1 voting disk(s).
2023/07/25 18:02:34 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:03:40 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:03:40 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:04:42 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:05:00 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@gncasnpl08358 ~]#

[root@gncasnpl08361 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= grid
    ORACLE_HOME=  /u01/app/19.19.0/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
  /u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/rootcrs_gncasnpl08361_2023-07-25_06-22-36PM.log
2023/07/25 18:22:42 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 18:22:43 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 18:22:43 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 18:22:47 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 18:22:48 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 18:22:49 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 18:22:50 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 18:22:51 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
2023/07/25 18:23:23 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:24:05 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:24:05 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:24:35 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:24:51 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
```

Install Oracle Database

```
[oracle@gncasnpl08358 ~]$ . db_profile
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$ xauth add gncasnpl08358/unix:11  MIT-MAGIC-COOKIE-1  61ffaebd4b2496351672b043d28ab8fc
[oracle@gncasnpl08358 dbhome_1]$ export DISPLAY=localhost:11.0
[oracle@gncasnpl08358 dbhome_1]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ ./runInstaller -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /u01/app/oraInventory/logs/InstallActions2023-07-25_11-42-20PM/installerPatchActions_2023-07-25_11-42-20PM.log
Launching Oracle Database Setup Wizard...

[FATAL] [INS-35071] Global database name cannot be left blank.
   CAUSE: The Global database name was left blank.
   ACTION: Specify a value for the Global database name.
```

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Conforme indicado no Grid Infrastructure, o swap o João Hoffman vai cuidar posteriormente e o NTP não é utilizado na versão 8.0

External Image
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Execução dos scripts:

```
[root@gncasnpl08358 ~]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME=  /u01/app/oracle/product/19.19.0/dbhome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.


[root@gncasnpl08361 app]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME=  /u01/app/oracle/product/19.19.0/dbhome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
```

- 2

INSTALL ORACLE RAC 19C IN LINUX – PRD

(Máquina física)

Criado em:26/07/2023
Última atualização em:28/07/2023
por:Alcides Souto

Você está aqui:

- [Main]
- [Getnet]
- INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

< Todos os tópicos
External ImageExternal Image
Projeto do PRM contempla 2 RACs, um RAC Ativo no DataCenter Sul e outro RAC no DataCenter Norte. Entre estes dois RACs haverá um Data Guard.

RAC Sul

- GNCASSPL08359
- GNCASSPL08360

RAC Norte

- GNCASNPL08358
- GNCASNPL08361

Este documento registra a instalação do RAC no Norte.

Entrega dos servidores

Servidores foram entregues pelo João Hoffman, tratam-se de máquinas físicas, sendo assim, revisamos diversos paramêtros de memoria, users e demais requisitos Oracle. Ao desenrolar a investigação, notamos que havia um StanAlone instalado em cada uma das máquinas.

Parâmetros de memória:

```
[root@gncasnpl08361 ~]# cat /etc/sysctl.conf
# System default settings live in /usr/lib/sysctl.d/00-system.conf.
# To override those settings, enter new settings here, or in an /etc/sysctl.d/<name>.conf file
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
# Controls IP packet forwarding
net.ipv4.ip_forward = 0
# Controls source route verification
net.ipv4.conf.default.rp_filter = 1
# Do not accept source routing
net.ipv4.conf.default.accept_source_route = 0
# Controls the System Request debugging functionality of the kernel
kernel.sysrq = 0
# Controls whether core dumps will append the PID to the core filename.
# Useful for debugging multi-threaded applications.
kernel.core_uses_pid = 1
# Controls the use of TCP syncookies
net.ipv4.tcp_syncookies = 1
# Disable netfilter on bridges.
net.bridge.bridge-nf-call-ip6tables = 0
net.bridge.bridge-nf-call-iptables = 0
net.bridge.bridge-nf-call-arptables = 0
# Controls the default maxmimum size of a mesage queue
kernel.msgmnb = 65536
# Controls the maximum size of a message, in bytes
kernel.msgmax = 65536
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmall = 4294967296
kernel.shmmni = 4096
kernel.sem = 250 32000 100 256
net.ipv4.ip_local_port_range = 9000 65500
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048586
kernel.shmmax = 270377232384
```

Todos os pacotes necessários, executar nos dois nós. Documentei apenas 1 deles.

```
[root@gncasnpl08358 ~]# dnf install bc binutils elfutils-libelf elfutils-libelf-devel fontconfig-devel glibc glibc-devel ksh libaio libaio-devel libgcc libnsl librdmacm-devel libstdc++ libstdc++-devel libX11 libXau libxcb libXi libXrender libXrender-devel libXtst make net-tools nfs-utils python3 python3-configshell python3-rtslib python3-six smartmontools sysstat targetcli unzip xorg-x11-apps chrony tmux xorg-x11-server-utils glibc.i686 glibc-devel.i686 libstdc++-devel.i686 libaio-devel.i686 libXext.i686 libXtst.i686 zlib-devel.i686 -y
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 54 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 43 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 47 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 26 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 28 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 44 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Package bc-1.07.1-5.el8.x86_64 is already installed.
Package binutils-2.30-119.el8.x86_64 is already installed.
Package elfutils-libelf-0.188-3.el8.x86_64 is already installed.
Package elfutils-libelf-devel-0.188-3.el8.x86_64 is already installed.
Package fontconfig-devel-2.13.1-4.el8.x86_64 is already installed.
```

```
Package glibc-2.28-225.el8.x86_64 is already installed.
Package glibc-devel-2.28-225.el8.x86_64 is already installed.
Package ksh-20120801-257.el8.x86_64 is already installed.
Package libaio-0.3.112-1.el8.x86_64 is already installed.
Package libaio-devel-0.3.112-1.el8.x86_64 is already installed.
Package libgcc-8.5.0-18.el8.x86_64 is already installed.
Package libnsl-2.28-225.el8.x86_64 is already installed.
Package rdma-core-devel-44.0-2.el8.1.x86_64 is already installed.
Package libstdc++-8.5.0-18.el8.x86_64 is already installed.
Package libstdc++-devel-8.5.0-18.el8.x86_64 is already installed.
Package libX11-1.6.8-5.el8.x86_64 is already installed.
Package libXau-1.0.9-3.el8.x86_64 is already installed.
Package libxcb-1.13.1-1.el8.x86_64 is already installed.
Package libXi-1.7.10-1.el8.x86_64 is already installed.
Package libXrender-0.9.10-7.el8.x86_64 is already installed.
Package libXrender-devel-0.9.10-7.el8.x86_64 is already installed.
Package libXtst-1.2.3-7.el8.x86_64 is already installed.
Package make-1:4.2.1-11.el8.x86_64 is already installed.
Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Package nfs-utils-1:2.3.3-59.el8.x86_64 is already installed.
Package python36-3.6.8-38.module+el8.5.0+12207+5c5719bc.x86_64 is already installed.
Package python3-six-1.11.0-8.el8.noarch is already installed.
Package smartmontools-1:7.1-1.el8.x86_64 is already installed.
Package sysstat-11.7.3-9.el8.x86_64 is already installed.
Package unzip-6.0-46.el8.x86_64 is already installed.
No match for argument: xorg-x11-apps
Package chrony-4.2-1.el8.x86_64 is already installed.
Package xorg-x11-server-utils-7.7-27.el8.x86_64 is already installed.
Error: Unable to find a match: xorg-x11-apps
[root@gncasnpl08358 ~]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 51 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 48 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 43 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 42 kB/s | 2.1 kB 00:00
Zabbix 6.0 37 kB/s | 2.1 kB 00:00
Package xorg-x11-xauth-1:1.0.9-12.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@gncasnpl08358 ~]# yum install xterm
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 43 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 47 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 43 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Dependencies resolved.
================================================================================================
Package Architecture Version Repository Size
================================================================================================
Installing:
xterm x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 529 k
Installing dependencies:
libXaw x86_64 1.0.13-10.el8 rhel-8-for-x86_64-appstream-rpms 194 k
xterm-resize x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 38 k
Installing weak dependencies:
xorg-x11-fonts-misc noarch 7.5-19.el8 rhel-8-for-x86_64-appstream-rpms 5.8 M
Transaction Summary
================================================================================================
Install 4 Packages
Total download size: 6.5 M
Installed size: 8.8 M
Is this ok [y/N]: y
Downloading Packages:
(1/4): xterm-331-1.el8_3.2.x86_64.rpm 5.3 MB/s | 529 kB 00:00
(2/4): xterm-resize-331-1.el8_3.2.x86_64.rpm 1.2 MB/s | 38 kB 00:00
(3/4): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm 26 MB/s | 5.8 MB 00:00
(4/4): libXaw-1.0.13-10.el8.x86_64.rpm 32 kB/s | 194 kB 00:06
------------------------------------------------------------------------------------------------
Total 1.1 MB/s | 6.5 MB 00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : xterm-resize-331-1.el8_3.2.x86_64 1/4
Installing : libXaw-1.0.13-10.el8.x86_64 2/4
Installing : xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Installing : xterm-331-1.el8_3.2.x86_64 4/4
```

Running scriptlet: xterm-331-1.el8_3.2.x86_64 4/4
Verifying : xorg-x11-fonts-misc-7.5-19.el8.noarch 1/4
Verifying : libXaw-1.0.13-10.el8.x86_64 2/4
Verifying : xterm-331-1.el8_3.2.x86_64 3/4
Verifying : xterm-resize-331-1.el8_3.2.x86_64 4/4
Installed products updated.
Installed:
libXaw-1.0.13-10.el8.x86_64 xorg-x11-fonts-misc-7.5-19.el8.noarch xterm-331-1.el8_3.2.x86_64 xterm-resize-331-1.el8_3.2.x86_64
Complete!
[root@gncasnpl08358 ~]#
Users oracle e grid já estavam configurados também, nos dois nos:

[root@gncasnpl08358 ~]# id oracle
uid=54340(oracle) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]# id grid
uid=54341(grid) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]#
Disable firewall, nos dois nos:

[root@gncasnpl08358 ~]# systemctl stop firewalld
[root@gncasnpl08358 ~]#
[root@gncasnpl08358 ~]# systemctl disable firewalld
Removed /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
Criação de profiles. Lembre que o nó 1 é PRMPRD1 e ASM1 e nó 2 PRMPRD2 e ASM2:

vi /home/oracle/db_profile
export ORACLE_BASE=/u01/app/oracle_19c
export ORACLE_HOME=/u01/app/oracle/product/19.19.0/dbhome_1
export ORACLE_SID=PRMCTG1
export OH=$ORACLE_HOME
export JAVA_HOME=/usr/bin/java
export GGS_HOME=/ggs
export PATH=.:/usr/bin/java/bin:/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/oracle/bin:$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib:$ORACLE_HOME/network/jlib
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$ORACLE_HOME/oracm/lib:/lib:/usr/lib:/usr/local/lib
export LIBPATH=/ggs:$ORACLE_HOME/lib:/usr/lib:$LIBPATH
chmod 775 /home/oracle/db_profile
chown oracle:oinstall /home/oracle/db_profile
vi /home/grid/grid_profile
export ORACLE_BASE=/u01/app/grid_19c
export ORACLE_HOME=/u01/app/19.19.0/grid
export ORACLE_SID=+ASM1;
export OH=$ORACLE_HOME
export PATH=$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
chmod 775 /home/grid/grid_profile
chown grid:oinstall /home/grid/grid_profile
Download e unzip dos pacotes, pode baixar nos 2 nós, mas os unzip da home do grid e oracle, somente no nó 01:

su - grid
. /home/grid/grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35037840_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35050341_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35004974_190000_Linux-x86-64.zip

oracleasm

João já entregou as máquinas com tudo instalado, sendo assim, vamos ter que desinstalar tudo e seguir exatamente como fizemos nos RACs de HML.

[root@gncasnpl08358 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
[root@gncasnpl08361 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04

DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13

Shutdown dos serviços e deinstall

```
[grid@gncasnpl08358 app]$ . /home/grid/grid_profile
[grid@gncasnpl08358 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08358' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08358' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[grid@gncasnpl08361 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08361' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08361' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[root@gncasnpl08358 ~]# . /home/grid/grid_profile
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]#
```

```
[root@gncasnpl08361 ~]# . /home/grid/grid_profile
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]#
[root@gncasnpl08358 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 ~]# oracleasm listdisks
[root@gncasnpl08361 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08361 ~]# oracleasm listdisks
```

Remoção dos binários:

```
[root@gncasnpl08358 ~]# cd /tmp
[root@gncasnpl08358 tmp]# rm -rf *
[root@gncasnpl08358 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08358 tmp]# cd /etc/oracle
[root@gncasnpl08358 oracle]# rm -rf *
[root@gncasnpl08358 oracle]# rm -rf /etc/oratab
[root@gncasnpl08358 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08358 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08358 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08358 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08358 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
```

```
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08358 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08358 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# cd /u01/
[root@gncasnpl08358 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 09:20 .
dr-xr-xr-x. 22 root root 4096 Jul 18 16:47 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:40 app
drwx------ 2 root root 16384 Jun 16 17:49 lost+found
[root@gncasnpl08358 u01]# rm -rf app
[root@gncasnpl08358 u01]#
[root@gncasnpl08361 ~]# cd /tmp
[root@gncasnpl08361 tmp]# rm -rf *
[root@gncasnpl08361 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08361 tmp]# cd /etc/oracle
[root@gncasnpl08361 oracle]# rm -rf *
[root@gncasnpl08361 oracle]# rm -rf /etc/oratab
[root@gncasnpl08361 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08361 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08361 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08361 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08361 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08361 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08361 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# cd /u01/
[root@gncasnpl08361 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 14:45 .
dr-xr-xr-x. 22 root root 4096 Jun 16 17:38 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:41 app
drwx------ 2 root root 16384 Jun 16 17:50 lost+found
[root@gncasnpl08361 u01]# rm -rf app
[root@gncasnpl08361 u01]#
```

Criação dos novos diretórios:

```
mkdir -p /u01/app/oracle/product/19.19.0/dbhome_1
mkdir -p /u01/app/19.19.0/grid
chown -R grid:oinstall /u01/app/19.19.0/
chown -R oracle:oinstall /u01/app/oracle/product/19.19.0
mkdir -p /u01/app/grid_19c
mkdir -p /u01/app/oracle_19c
chown -R grid:oinstall /u01/app/grid_19c/
chown -R oracle:oinstall /u01/app/oracle_19c/
chmod 775 /u01/app/oracle_19c/
chmod 775 /u01/app/grid_19c/
```

Formatação dos discos (que foram apagados previamente do ASM):

```
[root@gncasnpl08358 u01]# fdisk -l
Disk /dev/sda: 446.6 GiB, 479559942144 bytes, 936640512 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disklabel type: gpt
Disk identifier: EFD0B073-FC44-4EF7-B31F-0E3BA6ACFD10
Device Start End Sectors Size Type
/dev/sda1 2048 411647 409600 200M EFI System
/dev/sda2 411648 4605951 4194304 2G Microsoft basic data
/dev/sda3 4605952 6703103 2097152 1G Linux filesystem
/dev/sda4 6703104 936638463 929935360 443.4G Linux LVM
Disk /dev/sdb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
```

Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdd: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdd1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sde: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sde1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdf: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdf1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdg: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdh: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdh1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdi: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdi1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdm: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdm1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdn: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdn1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdo: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdo1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdp: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdp1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sds: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sds1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdt: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdt1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdw: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdw1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdx: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdx1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdy: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos

Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdy1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdz: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdz1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaa: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdaa1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdac: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdac1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdad: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdad1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdae: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdae1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdaf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdah1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdai: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdai1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdaj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdak: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdak1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdal: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdal1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdam: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdam1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdan: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdan1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdao: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdao1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdap: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdap1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdaq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdar: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdar1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdas: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdas1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdat: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdat1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdau: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdau1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdav: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos

Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdav1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaw: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdax: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdax1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sday: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sday1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdaz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdaz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdba: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdba1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdbd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbe: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbe1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdbf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdbg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbh: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdbh1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbi: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdbi1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdbj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdbk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdbl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbm: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdbm1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbn: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdbn1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbo: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdbo1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbp: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdbp1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbq: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdbq1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbr: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdbs: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdbs1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbt: 40 GiB, 42950983680 bytes, 83888640 sectors

Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdbt1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdbu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdbv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdby: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdby1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdca: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdca1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdcb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdcc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdcd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdce: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdce1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdcf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdcg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/rhel-root: 70 GiB, 75161927680 bytes, 146800640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-swap: 4 GiB, 4294967296 bytes, 8388608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-home: 369.4 GiB, 396667912192 bytes, 774742016 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/mpatha: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatha1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/vgu01-u01: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/vgu02-u02: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpaths: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpaths1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpatht: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatht1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathu: 1 TiB, 1099512545280 bytes, 2147485440 sectors

Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathy: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathy1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathaa: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaa1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathab1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/vg_ggs-ggs: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathac: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathac1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/mpathad: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathae: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathae1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathaf: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaf1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathah1 2048 2147485439 2147483392 1024G 83 Linux
[root@gncasnpl08358 u01]#
```

Output do multipath

```
[root@gncasnpl08358 u01]# multipath -ll
mpatha (360000970000597600012533030304431) dm-3 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:0 sdb 8:16 active ready running
|- 11:0:1:0 sdw 65:96 active ready running
|- 13:0:0:0 sdar 66:176 active ready running
`- 13:0:1:0 sdbm 68:0 active ready running
mpathaa (360000970000597600012533030313542) dm-20 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:20 sdv 65:80 active ready running
|- 11:0:1:20 sdaq 66:160 active ready running
|- 13:0:0:20 sdbl 67:240 active ready running
`- 13:0:1:20 sdcg 69:64 active ready running
mpathab (360000970000597600012533030303435) dm-22 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:3 sde 8:64 active ready running
|- 11:0:1:3 sdz 65:144 active ready running
|- 13:0:0:3 sdau 66:224 active ready running
`- 13:0:1:3 sdbp 68:48 active ready running
mpathac (360000970000597600012533030303436) dm-24 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:4 sdf 8:80 active ready running
|- 11:0:1:4 sdaa 65:160 active ready running
|- 13:0:0:4 sdav 66:240 active ready running
`- 13:0:1:4 sdbq 68:64 active ready running
mpathad (360000970000597600012533030313443) dm-25 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:5 sdg 8:96 active ready running
|- 11:0:1:5 sdab 65:176 active ready running
|- 13:0:0:5 sdaw 67:0 active ready running
`- 13:0:1:5 sdbr 68:80 active ready running
mpathae (360000970000597600012533030313444) dm-26 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:6 sdh 8:112 active ready running
|- 11:0:1:6 sdac 65:192 active ready running
|- 13:0:0:6 sdax 67:16 active ready running
`- 13:0:1:6 sdbs 68:96 active ready running
mpathaf (360000970000597600012533030313445) dm-27 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:7 sdi 8:128 active ready running
|- 11:0:1:7 sdad 65:208 active ready running
|- 13:0:0:7 sday 67:32 active ready running
`- 13:0:1:7 sdbt 68:112 active ready running
mpathag (360000970000597600012533030313446) dm-28 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:8 sdj 8:144 active ready running
|- 11:0:1:8 sdae 65:224 active ready running
|- 13:0:0:8 sdaz 67:48 active ready running
`- 13:0:1:8 sdbu 68:128 active ready running
mpathah (360000970000597600012533030313530) dm-29 EMC,SYMMETRIX
```

```
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:9 sdk 8:160 active ready running
|- 11:0:1:9 sdaf 65:240 active ready running
|- 13:0:0:9 sdba 67:64 active ready running
`- 13:0:1:9 sdbv 68:144 active ready running
mpathb (360000970000597600012533030304432) dm-4 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:1 sdc 8:32 active ready running
|- 11:0:1:1 sdx 65:112 active ready running
|- 13:0:0:1 sdas 66:192 active ready running
`- 13:0:1:1 sdbn 68:16 active ready running
mpathc (360000970000597600012533030304433) dm-19 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:2 sdd 8:48 active ready running
|- 11:0:1:2 sdy 65:128 active ready running
|- 13:0:0:2 sdat 66:208 active ready running
`- 13:0:1:2 sdbo 68:32 active ready running
mpathq (360000970000597600012533030313531) dm-6 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:10 sdl 8:176 active ready running
|- 11:0:1:10 sdag 66:0 active ready running
|- 13:0:0:10 sdbb 67:80 active ready running
`- 13:0:1:10 sdbw 68:160 active ready running
mpathr (360000970000597600012533030313532) dm-8 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:11 sdm 8:192 active ready running
|- 11:0:1:11 sdah 66:16 active ready running
|- 13:0:0:11 sdbc 67:96 active ready running
`- 13:0:1:11 sdbx 68:176 active ready running
mpaths (360000970000597600012533030313533) dm-11 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:12 sdn 8:208 active ready running
|- 11:0:1:12 sdai 66:32 active ready running
|- 13:0:0:12 sdbd 67:112 active ready running
`- 13:0:1:12 sdby 68:192 active ready running
mpatht (360000970000597600012533030313534) dm-12 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:13 sdo 8:224 active ready running
|- 11:0:1:13 sdaj 66:48 active ready running
|- 13:0:0:13 sdbe 67:128 active ready running
`- 13:0:1:13 sdbz 68:208 active ready running
mpathu (360000970000597600012533030313535) dm-13 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:14 sdp 8:240 active ready running
|- 11:0:1:14 sdak 66:64 active ready running
|- 13:0:0:14 sdbf 67:144 active ready running
`- 13:0:1:14 sdca 68:224 active ready running
mpathv (360000970000597600012533030313536) dm-14 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:15 sdq 65:0 active ready running
|- 11:0:1:15 sdal 66:80 active ready running
|- 13:0:0:15 sdbg 67:160 active ready running
`- 13:0:1:15 sdcb 68:240 active ready running
mpathw (360000970000597600012533030313537) dm-15 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:16 sdr 65:16 active ready running
|- 11:0:1:16 sdam 66:96 active ready running
|- 13:0:0:16 sdbh 67:176 active ready running
`- 13:0:1:16 sdcc 69:0 active ready running
mpathx (360000970000597600012533030313538) dm-16 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:17 sds 65:32 active ready running
|- 11:0:1:17 sdan 66:112 active ready running
|- 13:0:0:17 sdbi 67:192 active ready running
`- 13:0:1:17 sdcd 69:16 active ready running
mpathy (360000970000597600012533030313539) dm-17 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:18 sdt 65:48 active ready running
|- 11:0:1:18 sdao 66:128 active ready running
|- 13:0:0:18 sdbj 67:208 active ready running
`- 13:0:1:18 sdce 69:32 active ready running
mpathz (360000970000597600012533030313541) dm-18 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:19 sdu 65:64 active ready running
|- 11:0:1:19 sdap 66:144 active ready running
|- 13:0:0:19 sdbk 67:224 active ready running
`- 13:0:1:19 sdcf 69:48 active ready running
```

[root@gncasnpl08358 u01]#
Apagamos apenas os discos que foram utilizados na instalação standalone e reapresentamos ao ASM:

```
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathae1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00409259 s, 12.5 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaf1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00649853 s, 7.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathab1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00404717 s, 12.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathac1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00418631 s, 12.2 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathad1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.000105445 s, 486 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathq1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00391171 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathr1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00395635 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpaths1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00358853 s, 14.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpatht1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00451208 s, 11.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathu1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00439451 s, 11.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathv1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00373145 s, 13.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathw1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00607942 s, 8.4 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathx1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00389783 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathy1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00399422 s, 12.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathz1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00397336 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaa1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00370638 s, 13.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathag1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00364786 s, 14.0 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathah1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00387046 s, 13.2 MB/s
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD01 /dev/mapper/mpathae1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD02 /dev/mapper/mpathaf1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD01 /dev/mapper/mpathab1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD02 /dev/mapper/mpathac1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD01 /dev/mapper/mpathq1
Writing disk header: done
```

```
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD02 /dev/mapper/mpathr1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD03 /dev/mapper/mpaths1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD04 /dev/mapper/mpatht1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD05 /dev/mapper/mpathu1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD06 /dev/mapper/mpathv1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD07 /dev/mapper/mpathw1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD08 /dev/mapper/mpathx1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD09 /dev/mapper/mpathy1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD10 /dev/mapper/mpathz1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD11 /dev/mapper/mpathaa1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD12 /dev/mapper/mpathag1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD13 /dev/mapper/mpathah1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 u01]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```
Agora temos discos de ASM zerados e apresentados ao nó 01.
Podemos seguir com as questões de interfaces de rede. Consolidação dos IPs:

[External Image]
[External Image]
Resumo da configuração dos IPs:

GNCASNPL08358
IP Público: 180.104.147.12
IP Branca: 22.248.70.14
IP Backup: 22.240.70.12
Interconnect: 172.27.147.11
IP VIP: 180.104.147.15
Scan1: 180.104.147.19
Scan2: 180.104.147.20
Scan3: 180.104.147.21
GNCASNPL08361
IP Público: 180.104.147.14
IP Branca: 22.248.70.16
IP Backup: 22.240.70.15
Interconnect: 172.27.147.12
IP VIP: 180.104.147.18
Resolução das chaves de SSH:
NOTA: o documento de RAC de HML está completo a parte do SSH.

```
passwd oracle
passwd grid
*** usar uma senha padrão
./sshUserSetup.sh -user grid -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
./sshUserSetup.sh -user oracle -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
chmod 600 /home/grid/.ssh/config
```

chmod 600 /home/oracle/.ssh/config
Configuração do /etc/hosts:

```
### Install RAC Oracle
180.104.147.12 GNCASNPL08358 GNCASNPL08358.getnet.local
22.240.70.12 GNCASNPL08358.bkp GNCASNPL08358.bkp.getnet.local
22.248.70.14 GNCASNPL08358.wt GNCASNPL08358.wt.getnet.local
180.104.147.15 GNCASNPL08358-vip GNCASNPL08358-vip.getnet.local
180.104.147.14 GNCASNPL08361 GNCASNPL08361.getnet.local
22.240.70.15 GNCASNPL08361.bkp GNCASNPL08361.bkp.getnet.local
22.248.70.16 GNCASNPL08361.wt GNCASNPL08361.wt.getnet.local
180.104.147.18 GNCASNPL08361-vip GNCASNPL08361-vip.getnet.local
```

Download do patch que corrige o SSH:

```
su - grid
. grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_Linux-x86-64.zip
unzip p30159782_195000OCWRU_Linux-x86-64.zip
```

Install Grid Infrastructure

```
[grid@gncasnpl08358 ~]$ . grid_profile
[grid@gncasnpl08358 ~]$ cd $ORACLE_HOME
[grid@gncasnpl08358 grid]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[grid@gncasnpl08358 grid]$ export DISPLAY=localhost:11.0
[grid@gncasnpl08358 grid]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 grid]$ ./gridSetup.sh -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /tmp/GridSetupActions2023-07-20_04-46-39PM/installerPatchActions_2023-07-20_04-46-39PM.log
Launching Oracle Grid Infrastructure Setup Wizard...
[FATAL] [INS-40426] Grid installation option has not been specified.
ACTION: Specify the valid installation option.
[grid@gncasnpl08358 grid]$ ./gridSetup.sh
```

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Adicionar o segundo nó do cluster:
Clique em "Add…"

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Como trata-se de máquinas físicas, há mais interfaces de redes configuradas, como da idrac, por exemplo, lembre-se de utilizar a interface publica para o public e interconnect para ASM & Private:

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NOTA: O erro abaixo está relacionado a interface do IDrac, conversei com o João HOffman, ele me pediu para ignorar:

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External Image
NOTA: lembre-se de ajustar o discovery path:

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Escolhemos apenas discos do +DATA:

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External Image
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External Image
Observações:
Node Connectivity é a respeito do Idrac, o swap eu combinei do João Hoffman ajustar posteriormente dado a emergencia do projeto e os demais são irrelevantes:

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Execução de scripts:

```
[root@gncasnpl08358 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08361 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08358 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
```

The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/rootcrs_gncasnpl08358_2023-07-25_05-58-37PM.log
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 17:58:46 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 17:58:47 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 17:58:56 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 17:58:59 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 17:59:07 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 17:59:14 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 17:59:14 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/07/25 17:59:34 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 17:59:38 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 17:59:42 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 17:59:45 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
ASM has been created and started successfully.
[DBT-30001] Disk groups created successfully. Check /u01/app/grid_19c/cfgtoollogs/asmca/asmca-230725PM060015.log for details.
2023/07/25 18:01:21 CLSRSC-482: Running command: '/u01/app/19.19.0/grid/bin/ocrconfig -upgrade grid oinstall'
CRS-4256: Updating the profile
Successful addition of voting disk 25590c0c233d4f26bf21c16dcb0516a4.
Successfully replaced voting disk group with +DATA_PRM84PRD.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
STATE File Universal Id File Name Disk group
-- ----- ----------------- --------- ---------
1. ONLINE 25590c0c233d4f26bf21c16dcb0516a4 (/dev/oracleasm/disks/DATA_PRM84PRD01) [DATA_PRM84PRD]
Located 1 voting disk(s).
2023/07/25 18:02:34 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:03:40 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:03:40 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:04:42 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:05:00 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@gncasnpl08358 ~]#
[root@gncasnpl08361 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/rootcrs_gncasnpl08361_2023-07-25_06-22-36PM.log
2023/07/25 18:22:42 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 18:22:43 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 18:22:43 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 18:22:47 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 18:22:48 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 18:22:49 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 18:22:50 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 18:22:51 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
2023/07/25 18:23:23 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:24:05 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:24:05 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:24:35 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:24:51 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded

Install Oracle Database

[oracle@gncasnpl08358 ~]$. db_profile
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[oracle@gncasnpl08358 dbhome_1]$ export DISPLAY=localhost:11.0
[oracle@gncasnpl08358 dbhome_1]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ ./runInstaller -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /u01/app/oraInventory/logs/InstallActions2023-07-25_11-42-20PM/installerPatchActions_2023-07-25_11-42-20PM.log
Launching Oracle Database Setup Wizard...
[FATAL] [INS-35071] Global database name cannot be left blank.
CAUSE: The Global database name was left blank.
ACTION: Specify a value for the Global database name.
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Conforme indicado no Grid Infrastructure, o swap o João Hoffman vai cuidar posteriormente e o NTP não é utilizado na versão 8.0

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Execução dos scripts:

[root@gncasnpl08358 ~]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@gncasnpl08361 app]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
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ASMCA

Criação dos Diskgroups faltantes com user Grid.

```
[grid@gncasnpl08358 ~]$ . grid_profile
[grid@gncasnpl08358 ~]$  xauth add gncasnpl08358/unix:10  MIT-MAGIC-COOKIE-1  62fdb29b8a741ad74f6536662ad2b3d0
[grid@gncasnpl08358 ~]$  export DISPLAY=localhost:10.0
[grid@gncasnpl08358 ~]$  export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 ~]$  asmca
```

- 2

INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

Criado em:26/07/2023
Última atualização em:28/07/2023
por:Alcides Souto

Você está aqui:
- Main
- Getnet
- INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

< Todos os tópicos
External ImageExternal Image
Projeto do PRM contempla 2 RACs, um RAC Ativo no DataCenter Sul e outro RAC no DataCenter Norte. Entre estes dois RACs haverá um Data Guard.

RAC Sul

- GNCASSPL08359
- GNCASSPL08360

RAC Norte

- GNCASNPL08358
- GNCASNPL08361

Este documento registra a instalação do RAC no Norte.

Entrega dos servidores

Servidores foram entregues pelo João Hoffman, tratam-se de máquinas físicas, sendo assim, revisamos diversos paramêtros de memoria, users e demais requisitos Oracle. Ao desenrolar a investigação, notamos que havia um StanAlone instalado em cada uma das máquinas.

Parâmetros de memória:

```
[root@gncasnpl08361 ~]# cat /etc/sysctl.conf
# System default settings live in /usr/lib/sysctl.d/00-system.conf.
# To override those settings, enter new settings here, or in an /etc/sysctl.d/<name>.conf file
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
# Controls IP packet forwarding
net.ipv4.ip_forward = 0
# Controls source route verification
net.ipv4.conf.default.rp_filter = 1
# Do not accept source routing
net.ipv4.conf.default.accept_source_route = 0
# Controls the System Request debugging functionality of the kernel
kernel.sysrq = 0
# Controls whether core dumps will append the PID to the core filename.
# Useful for debugging multi-threaded applications.
kernel.core_uses_pid = 1
# Controls the use of TCP syncookies
net.ipv4.tcp_syncookies = 1
# Disable netfilter on bridges.
net.bridge.bridge-nf-call-ip6tables = 0
net.bridge.bridge-nf-call-iptables = 0
net.bridge.bridge-nf-call-arptables = 0
# Controls the default maxmimum size of a mesage queue
kernel.msgmnb = 65536
# Controls the maximum size of a message, in bytes
kernel.msgmax = 65536
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmall = 4294967296
kernel.shmmni = 4096
kernel.sem = 250 32000 100 256
net.ipv4.ip_local_port_range = 9000 65500
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048586
kernel.shmmax = 270377232384
```

Todos os pacotes necessários, executar nos dois nós. Documentei apenas 1 deles.

```
[root@gncasnpl08358 ~]# dnf install bc binutils elfutils-libelf elfutils-libelf-devel fontconfig-devel glibc glibc-devel ksh libaio libaio-devel libgcc libnsl librdmacm-devel libstdc++ libstdc++-devel libX11 libXau libxcb libXi libXrender libXrender-devel libXtst make net-tools nfs-utils python3 python3-configshell python3-rtslib python3-six smartmontools sysstat targetcli unzip xorg-x11-apps chrony tmux xorg-x11-server-utils glibc.i686 glibc-devel.i686 libstdc++-devel.i686 libaio-devel.i686 libXext.i686 libXtst.i686 zlib-devel.i686 -y
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 54 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 43 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 47 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 26 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 28 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 44 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Package bc-1.07.1-5.el8.x86_64 is already installed.
Package binutils-2.30-119.el8.x86_64 is already installed.
Package elfutils-libelf-0.188-3.el8.x86_64 is already installed.
Package elfutils-libelf-devel-0.188-3.el8.x86_64 is already installed.
Package fontconfig-devel-2.13.1-4.el8.x86_64 is already installed.
Package glibc-2.28-225.el8.x86_64 is already installed.
Package glibc-devel-2.28-225.el8.x86_64 is already installed.
Package ksh-20120801-257.el8.x86_64 is already installed.
Package libaio-0.3.112-1.el8.x86_64 is already installed.
Package libaio-devel-0.3.112-1.el8.x86_64 is already installed.
Package libgcc-8.5.0-18.el8.x86_64 is already installed.
Package libnsl-2.28-225.el8.x86_64 is already installed.
Package rdma-core-devel-44.0-2.el8.1.x86_64 is already installed.
Package libstdc++-8.5.0-18.el8.x86_64 is already installed.
Package libstdc++-devel-8.5.0-18.el8.x86_64 is already installed.
Package libX11-1.6.8-5.el8.x86_64 is already installed.
Package libXau-1.0.9-3.el8.x86_64 is already installed.
Package libxcb-1.13.1-1.el8.x86_64 is already installed.
Package libXi-1.7.10-1.el8.x86_64 is already installed.
Package libXrender-0.9.10-7.el8.x86_64 is already installed.
Package libXrender-devel-0.9.10-7.el8.x86_64 is already installed.
```

Package libXtst-1.2.3-7.el8.x86_64 is already installed.
Package make-1:4.2.1-11.el8.x86_64 is already installed.
Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Package nfs-utils-1:2.3.3-59.el8.x86_64 is already installed.
Package python36-3.6.8-38.module+el8.5.0+12207+5c5719bc.x86_64 is already installed.
Package python3-six-1.11.0-8.el8.noarch is already installed.
Package smartmontools-1:7.1-1.el8.x86_64 is already installed.
Package sysstat-11.7.3-9.el8.x86_64 is already installed.
Package unzip-6.0-46.el8.x86_64 is already installed.
No match for argument: xorg-x11-apps
Package chrony-4.2-1.el8.x86_64 is already installed.
Package xorg-x11-server-utils-7.7-27.el8.x86_64 is already installed.
Error: Unable to find a match: xorg-x11-apps
[root@gncasnpl08358 ~]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 51 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 48 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 43 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 42 kB/s | 2.1 kB 00:00
Zabbix 6.0 37 kB/s | 2.1 kB 00:00
Package xorg-x11-xauth-1:1.0.9-12.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@gncasnpl08358 ~]# yum install xterm
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 43 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 47 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 43 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Dependencies resolved.
==
Package Architecture Version Repository Size
==
Installing:
xterm x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 529 k
Installing dependencies:
libXaw x86_64 1.0.13-10.el8 rhel-8-for-x86_64-appstream-rpms 194 k
xterm-resize x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 38 k
Installing weak dependencies:
xorg-x11-fonts-misc noarch 7.5-19.el8 rhel-8-for-x86_64-appstream-rpms 5.8 M
Transaction Summary
==
Install 4 Packages
Total download size: 6.5 M
Installed size: 8.8 M
Is this ok [y/N]: y
Downloading Packages:
(1/4): xterm-331-1.el8_3.2.x86_64.rpm 5.3 MB/s | 529 kB 00:00
(2/4): xterm-resize-331-1.el8_3.2.x86_64.rpm 1.2 MB/s | 38 kB 00:00
(3/4): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm 26 MB/s | 5.8 MB 00:00
(4/4): libXaw-1.0.13-10.el8.x86_64.rpm 32 kB/s | 194 kB 00:06

Total 1.1 MB/s | 6.5 MB 00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : xterm-resize-331-1.el8_3.2.x86_64 1/4
Installing : libXaw-1.0.13-10.el8.x86_64 2/4
Installing : xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Installing : xterm-331-1.el8_3.2.x86_64 4/4
Running scriptlet: xterm-331-1.el8_3.2.x86_64 4/4
Verifying : xorg-x11-fonts-misc-7.5-19.el8.noarch 1/4
Verifying : libXaw-1.0.13-10.el8.x86_64 2/4
Verifying : xterm-331-1.el8_3.2.x86_64 3/4
Verifying : xterm-resize-331-1.el8_3.2.x86_64 4/4
Installed products updated.
Installed:
libXaw-1.0.13-10.el8.x86_64 xorg-x11-fonts-misc-7.5-19.el8.noarch xterm-331-1.el8_3.2.x86_64 xterm-resize-331-1.el8_3.2.x86_64
Complete!
[root@gncasnpl08358 ~]#
Users oracle e grid já estavam configurados também, nos dois nos:

[root@gncasnpl08358 ~]# id oracle
uid=54340(oracle) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]# id grid
uid=54341(grid) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)

[root@gncasnpl08358 ~]#
Disable firewall, nos dois nos:

```
[root@gncasnpl08358 ~]# systemctl stop firewalld
[root@gncasnpl08358 ~]#
[root@gncasnpl08358 ~]# systemctl disable firewalld
Removed /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
```

Criação de profiles. Lembre que o nó 1 é PRMPRD1 e ASM1 e nó 2 PRMPRD2 e ASM2:

```
vi /home/oracle/db_profile
export ORACLE_BASE=/u01/app/oracle_19c
export ORACLE_HOME=/u01/app/oracle/product/19.19.0/dbhome_1
export ORACLE_SID=PRMCTG1
export OH=$ORACLE_HOME
export JAVA_HOME=/usr/bin/java
export GGS_HOME=/ggs
export PATH=.:/usr/bin/java/bin:/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/oracle/bin:$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib:$ORACLE_HOME/network/jlib
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$ORACLE_HOME/oracm/lib:/lib:/usr/lib:/usr/local/lib
export LIBPATH=/ggs:$ORACLE_HOME/lib:/usr/lib:$LIBPATH
chmod 775 /home/oracle/db_profile
chown oracle:oinstall /home/oracle/db_profile
vi /home/grid/grid_profile
export ORACLE_BASE=/u01/app/grid_19c
export ORACLE_HOME=/u01/app/19.19.0/grid
export ORACLE_SID=+ASM1;
export OH=$ORACLE_HOME
export PATH=$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
chmod 775 /home/grid/grid_profile
chown grid:oinstall /home/grid/grid_profile
```

Download e unzip dos pacotes, pode baixar nos 2 nós, mas os unzip da home do grid e oracle, somente no nó 01:

```
su - grid
. /home/grid/grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35037840_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35050341_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35004974_190000_Linux-x86-64.zip
```

oracleasm

João já entregou as máquinas com tudo instalado, sendo assim, vamos ter que desinstalar tudo e seguir exatamente como fizemos nos RACs de HML.

```
[root@gncasnpl08358 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
[root@gncasnpl08361 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```

Shutdown dos serviços e deinstall

```
[grid@gncasnpl08358 app]$ . /home/grid/grid_profile
[grid@gncasnpl08358 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08358'
```

```
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08358' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08358' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[grid@gncasnpl08361 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08361' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08361' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[root@gncasnpl08358 ~]# . /home/grid/grid_profile
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]#
[root@gncasnpl08361 ~]# . /home/grid/grid_profile
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
```

```
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]#
[root@gncasnpl08358 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 ~]# oracleasm listdisks
[root@gncasnpl08361 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08361 ~]# oracleasm listdisks
```

Remoção dos binários:

```
[root@gncasnpl08358 ~]# cd /tmp
[root@gncasnpl08358 tmp]# rm -rf *
[root@gncasnpl08358 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08358 tmp]# cd /etc/oracle
[root@gncasnpl08358 oracle]# rm -rf *
[root@gncasnpl08358 oracle]# rm -rf /etc/oratab
[root@gncasnpl08358 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08358 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08358 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08358 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08358 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08358 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08358 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# cd /u01/
[root@gncasnpl08358 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 09:20 .
dr-xr-xr-x. 22 root root 4096 Jul 18 16:47 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:40 app
drwx------ 2 root root 16384 Jun 16 17:49 lost+found
[root@gncasnpl08358 u01]# rm -rf app
```

```
[root@gncasnpl08358 u01]#
[root@gncasnpl08361 ~]# cd /tmp
[root@gncasnpl08361 tmp]# rm -rf *
[root@gncasnpl08361 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08361 tmp]# cd /etc/oracle
[root@gncasnpl08361 oracle]# rm -rf *
[root@gncasnpl08361 oracle]# rm -rf /etc/oratab
[root@gncasnpl08361 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08361 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08361 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08361 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08361 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08361 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08361 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# cd /u01/
[root@gncasnpl08361 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 14:45 .
dr-xr-xr-x. 22 root root 4096 Jun 16 17:38 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:41 app
drwx------ 2 root root 16384 Jun 16 17:50 lost+found
[root@gncasnpl08361 u01]# rm -rf app
[root@gncasnpl08361 u01]#
```

Criação dos novos diretórios:

```
mkdir -p /u01/app/oracle/product/19.19.0/dbhome_1
mkdir -p /u01/app/19.19.0/grid
chown -R grid:oinstall /u01/app/19.19.0/
chown -R oracle:oinstall /u01/app/oracle/product/19.19.0
mkdir -p /u01/app/grid_19c
mkdir -p /u01/app/oracle_19c
chown -R grid:oinstall /u01/app/grid_19c/
chown -R oracle:oinstall /u01/app/oracle_19c/
chmod 775 /u01/app/oracle_19c/
chmod 775 /u01/app/grid_19c/
```

Formatação dos discos (que foram apagados previamente do ASM):

```
[root@gncasnpl08358 u01]# fdisk -l
Disk /dev/sda: 446.6 GiB, 479559942144 bytes, 936640512 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disklabel type: gpt
Disk identifier: EFD0B073-FC44-4EF7-B31F-0E3BA6ACFD10
Device Start End Sectors Size Type
/dev/sda1 2048 411647 409600 200M EFI System
/dev/sda2 411648 4605951 4194304 2G Microsoft basic data
/dev/sda3 4605952 6703103 2097152 1G Linux filesystem
/dev/sda4 6703104 936638463 929935360 443.4G Linux LVM
Disk /dev/sdb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdd: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
```

Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdd1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sde: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sde1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdf: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdf1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdg: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdh: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdh1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdi: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdi1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdm: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdm1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdn: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdn1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdo: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdo1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdp: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdp1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sds: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sds1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdt: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdt1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdw: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdw1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdx: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdx1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdy: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdy1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdz: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdz1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaa: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos

Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdaa1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdac: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdac1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdad: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdad1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdae: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdae1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdaf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdah1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdai: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdai1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdaj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdak: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdak1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdal: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdal1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdam: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdam1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdan: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdan1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdao: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdao1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdap: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdap1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdaq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdar: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdar1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdas: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdas1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdat: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdat1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdau: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdau1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdav: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdav1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaw: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdax: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdax1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sday: 40 GiB, 42950983680 bytes, 83888640 sectors

Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sday1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdaz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdaz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdba: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdba1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdbd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbe: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbe1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdbf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdbg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbh: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdbh1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbi: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdbi1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdbj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdbk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdbl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbm: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdbm1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbn: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdbn1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbo: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdbo1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbp: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdbp1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbq: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdbq1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbr: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdbs: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdbs1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbt: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdbt1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdbu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbv: 1 TiB, 1099512545280 bytes, 2147485440 sectors

Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdbv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdby: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdby1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdca: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdca1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdcb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdcc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdcd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdce: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdce1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdcf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdcg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/rhel-root: 70 GiB, 75161927680 bytes, 146800640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-swap: 4 GiB, 4294967296 bytes, 8388608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-home: 369.4 GiB, 396667912192 bytes, 774742016 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/mpatha: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatha1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/vgu01-u01: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/vgu02-u02: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpaths: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpaths1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpatht: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatht1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathw: 1 TiB, 1099512545280 bytes, 2147485440 sectors

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Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathy: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathy1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathaa: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaa1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathab1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/vg_ggs-ggs: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathac: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathac1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/mpathad: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathae: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathae1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathaf: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaf1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathah1 2048 2147485439 2147483392 1024G 83 Linux
[root@gncasnpl08358 u01]#
Output do multipath

[root@gncasnpl08358 u01]# multipath -ll
mpatha (360000970000597600012533030304431) dm-3 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:0 sdb 8:16 active ready running
|- 11:0:1:0 sdw 65:96 active ready running
|- 13:0:0:0 sdar 66:176 active ready running
`- 13:0:1:0 sdbm 68:0 active ready running
mpathaa (360000970000597600012533030313542) dm-20 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:20 sdv 65:80 active ready running
|- 11:0:1:20 sdaq 66:160 active ready running
|- 13:0:0:20 sdbl 67:240 active ready running
`- 13:0:1:20 sdcg 69:64 active ready running
mpathab (360000970000597600012533030303435) dm-22 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:3 sde 8:64 active ready running
|- 11:0:1:3 sdz 65:144 active ready running
|- 13:0:0:3 sdau 66:224 active ready running
`- 13:0:1:3 sdbp 68:48 active ready running
mpathac (360000970000597600012533030303436) dm-24 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:4 sdf 8:80 active ready running
|- 11:0:1:4 sdaa 65:160 active ready running
|- 13:0:0:4 sdav 66:240 active ready running
`- 13:0:1:4 sdbq 68:64 active ready running
mpathad (360000970000597600012533030313443) dm-25 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:5 sdg 8:96 active ready running
|- 11:0:1:5 sdab 65:176 active ready running
|- 13:0:0:5 sdaw 67:0 active ready running
`- 13:0:1:5 sdbr 68:80 active ready running
mpathae (360000970000597600012533030313444) dm-26 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:6 sdh 8:112 active ready running
|- 11:0:1:6 sdac 65:192 active ready running
|- 13:0:0:6 sdax 67:16 active ready running
`- 13:0:1:6 sdbs 68:96 active ready running
mpathaf (360000970000597600012533030313445) dm-27 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:7 sdi 8:128 active ready running
|- 11:0:1:7 sdad 65:208 active ready running
|- 13:0:0:7 sday 67:32 active ready running
`- 13:0:1:7 sdbt 68:112 active ready running
mpathag (360000970000597600012533030313446) dm-28 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:8 sdj 8:144 active ready running
|- 11:0:1:8 sdae 65:224 active ready running
|- 13:0:0:8 sdaz 67:48 active ready running
`- 13:0:1:8 sdbu 68:128 active ready running
mpathah (360000970000597600012533030313530) dm-29 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:9 sdk 8:160 active ready running
|- 11:0:1:9 sdaf 65:240 active ready running
|- 13:0:0:9 sdba 67:64 active ready running
`- 13:0:1:9 sdbv 68:144 active ready running
mpathb (360000970000597600012533030304432) dm-4 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:1 sdc 8:32 active ready running
|- 11:0:1:1 sdx 65:112 active ready running
|- 13:0:0:1 sdas 66:192 active ready running
`- 13:0:1:1 sdbn 68:16 active ready running
mpathc (360000970000597600012533030304433) dm-19 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active

```
|- 11:0:0:2 sdd 8:48 active ready running
|- 11:0:1:2 sdy 65:128 active ready running
|- 13:0:0:2 sdat 66:208 active ready running
`- 13:0:1:2 sdbo 68:32 active ready running
mpathq (360000970000597600012533030313531) dm-6 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:10 sdl 8:176 active ready running
|- 11:0:1:10 sdag 66:0 active ready running
|- 13:0:0:10 sdbb 67:80 active ready running
`- 13:0:1:10 sdbw 68:160 active ready running
mpathr (360000970000597600012533030313532) dm-8 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:11 sdm 8:192 active ready running
|- 11:0:1:11 sdah 66:16 active ready running
|- 13:0:0:11 sdbc 67:96 active ready running
`- 13:0:1:11 sdbx 68:176 active ready running
mpaths (360000970000597600012533030313533) dm-11 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:12 sdn 8:208 active ready running
|- 11:0:1:12 sdai 66:32 active ready running
|- 13:0:0:12 sdbd 67:112 active ready running
`- 13:0:1:12 sdby 68:192 active ready running
mpatht (360000970000597600012533030313534) dm-12 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:13 sdo 8:224 active ready running
|- 11:0:1:13 sdaj 66:48 active ready running
|- 13:0:0:13 sdbe 67:128 active ready running
`- 13:0:1:13 sdbz 68:208 active ready running
mpathu (360000970000597600012533030313535) dm-13 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:14 sdp 8:240 active ready running
|- 11:0:1:14 sdak 66:64 active ready running
|- 13:0:0:14 sdbf 67:144 active ready running
`- 13:0:1:14 sdca 68:224 active ready running
mpathv (360000970000597600012533030313536) dm-14 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:15 sdq 65:0 active ready running
|- 11:0:1:15 sdal 66:80 active ready running
|- 13:0:0:15 sdbg 67:160 active ready running
`- 13:0:1:15 sdcb 68:240 active ready running
mpathw (360000970000597600012533030313537) dm-15 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:16 sdr 65:16 active ready running
|- 11:0:1:16 sdam 66:96 active ready running
|- 13:0:0:16 sdbh 67:176 active ready running
`- 13:0:1:16 sdcc 69:0 active ready running
mpathx (360000970000597600012533030313538) dm-16 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:17 sds 65:32 active ready running
|- 11:0:1:17 sdan 66:112 active ready running
|- 13:0:0:17 sdbi 67:192 active ready running
`- 13:0:1:17 sdcd 69:16 active ready running
mpathy (360000970000597600012533030313539) dm-17 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:18 sdt 65:48 active ready running
|- 11:0:1:18 sdao 66:128 active ready running
|- 13:0:0:18 sdbj 67:208 active ready running
`- 13:0:1:18 sdce 69:32 active ready running
mpathz (360000970000597600012533030313541) dm-18 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:19 sdu 65:64 active ready running
|- 11:0:1:19 sdap 66:144 active ready running
|- 13:0:0:19 sdbk 67:224 active ready running
`- 13:0:1:19 sdcf 69:48 active ready running
[root@gncasnpl08358 u01]#
```

Apagamos apenas os discos que foram utilizados na instalação standalone e reapresentamos ao ASM:

```
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathae1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00409259 s, 12.5 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaf1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00649853 s, 7.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathab1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00404717 s, 12.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathac1 bs=512 count=100
```

```
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00418631 s, 12.2 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathad1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.000105445 s, 486 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathq1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00391171 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathr1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00395635 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpaths1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00358853 s, 14.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpatht1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00451208 s, 11.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathu1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00439451 s, 11.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathv1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00373145 s, 13.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathw1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00607942 s, 8.4 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathx1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00389783 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathy1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00399422 s, 12.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathz1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00397336 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaa1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00370638 s, 13.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathag1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00364786 s, 14.0 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathah1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00387046 s, 13.2 MB/s
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD01 /dev/mapper/mpathae1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD02 /dev/mapper/mpathaf1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD01 /dev/mapper/mpathab1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD02 /dev/mapper/mpathac1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD01 /dev/mapper/mpathq1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD02 /dev/mapper/mpathr1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD03 /dev/mapper/mpaths1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD04 /dev/mapper/mpatht1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD05 /dev/mapper/mpathu1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD06 /dev/mapper/mpathv1
Writing disk header: done
Instantiating disk: done
```

[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD07 /dev/mapper/mpathw1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD08 /dev/mapper/mpathx1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD09 /dev/mapper/mpathy1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD10 /dev/mapper/mpathz1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD11 /dev/mapper/mpathaa1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD12 /dev/mapper/mpathag1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD13 /dev/mapper/mpathah1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 u01]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
Agora temos discos de ASM zerados e apresentados ao nó 01.
Podemos seguir com as questões de interfaces de rede. Consolidação dos IPs:

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Resumo da configuração dos IPs:

GNCASNPL08358
IP Público: 180.104.147.12
IP Branca: 22.248.70.14
IP Backup: 22.240.70.12
Interconnect: 172.27.147.11
IP VIP: 180.104.147.15
Scan1: 180.104.147.19
Scan2: 180.104.147.20
Scan3: 180.104.147.21
GNCASNPL08361
IP Público: 180.104.147.14
IP Branca: 22.248.70.16
IP Backup: 22.240.70.15
Interconnect: 172.27.147.12
IP VIP: 180.104.147.18
Resolução das chaves de SSH:
NOTA: o documento de RAC de HML está completo a parte do SSH.

passwd oracle
passwd grid
*** usar uma senha padrão
./sshUserSetup.sh -user grid -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
./sshUserSetup.sh -user oracle -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
chmod 600 /home/grid/.ssh/config
chmod 600 /home/oracle/.ssh/config
Configuração do /etc/hosts:

Install RAC Oracle
180.104.147.12 GNCASNPL08358 GNCASNPL08358.getnet.local
22.240.70.12 GNCASNPL08358.bkp GNCASNPL08358.bkp.getnet.local
22.248.70.14 GNCASNPL08358.wt GNCASNPL08358.wt.getnet.local
180.104.147.15 GNCASNPL08358-vip GNCASNPL08358-vip.getnet.local
180.104.147.14 GNCASNPL08361 GNCASNPL08361.getnet.local
22.240.70.15 GNCASNPL08361.bkp GNCASNPL08361.bkp.getnet.local
22.248.70.16 GNCASNPL08361.wt GNCASNPL08361.wt.getnet.local
180.104.147.18 GNCASNPL08361-vip GNCASNPL08361-vip.getnet.local
Download do patch que corrige o SSH:

su - grid
. grid_profile

cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_Linux-x86-64.zip
unzip p30159782_195000OCWRU_Linux-x86-64.zip
Install Grid Infrastructure

[grid@gncasnpl08358 ~]$. grid_profile
[grid@gncasnpl08358 ~]$ cd $ORACLE_HOME
[grid@gncasnpl08358 grid]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[grid@gncasnpl08358 grid]$ export DISPLAY=localhost:11.0
[grid@gncasnpl08358 grid]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 grid]$ ./gridSetup.sh -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /tmp/GridSetupActions2023-07-20_04-46-39PM/installerPatchActions_2023-07-20_04-46-39PM.log
Launching Oracle Grid Infrastructure Setup Wizard...
[FATAL] [INS-40426] Grid installation option has not been specified.
ACTION: Specify the valid installation option.
[grid@gncasnpl08358 grid]$ ./gridSetup.sh
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Adicionar o segundo nó do cluster:
Clique em “Add…”

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Como trata-se de máquinas físicas, há mais interfaces de redes configuradas, como da idrac, por exemplo, lembre-se de utilizar a interface publica para o public e interconnect para ASM & Private:

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NOTA: O erro abaixo está relacionado a interface do IDrac, conversei com o João HOffman, ele me pediu para ignorar:

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NOTA: lembre-se de ajustar o discovery path:

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Escolhemos apenas discos do +DATA:

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Observações:
Node Connectivity é a respeito do Idrac, o swap eu combinei do João Hoffman ajustar posteriormente dado a emergencia do projeto e os demais são irrelevantes:

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Execução de scripts:

[root@gncasnpl08358 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08361 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08358 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/rootcrs_gncasnpl08358_2023-07-25_05-58-37PM.log
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 17:58:46 CLSRSC-363: User ignored prerequisites during installation

2023/07/25 17:58:46 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 17:58:47 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 17:58:56 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 17:58:59 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 17:59:07 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 17:59:14 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 17:59:14 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/07/25 17:59:34 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 17:59:38 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 17:59:42 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 17:59:45 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
ASM has been created and started successfully.
[DBT-30001] Disk groups created successfully. Check /u01/app/grid_19c/cfgtoollogs/asmca/asmca-230725PM060015.log for details.
2023/07/25 18:01:21 CLSRSC-482: Running command: '/u01/app/19.19.0/grid/bin/ocrconfig -upgrade grid oinstall'
CRS-4256: Updating the profile
Successful addition of voting disk 25590c0c233d4f26bf21c16dcb0516a4.
Successfully replaced voting disk group with +DATA_PRM84PRD.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
STATE File Universal Id File Name Disk group
-- ----- ----------------- --------- ---------
1. ONLINE 25590c0c233d4f26bf21c16dcb0516a4 (/dev/oracleasm/disks/DATA_PRM84PRD01) [DATA_PRM84PRD]
Located 1 voting disk(s).
2023/07/25 18:02:34 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:03:40 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:03:40 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:04:42 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:05:00 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@gncasnpl08358 ~]#
[root@gncasnpl08361 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/rootcrs_gncasnpl08361_2023-07-25_06-22-36PM.log
2023/07/25 18:22:42 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 18:22:43 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 18:22:43 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 18:22:47 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 18:22:48 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 18:22:49 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 18:22:50 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 18:22:51 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
2023/07/25 18:23:23 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:24:05 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:24:05 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:24:35 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:24:51 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded

Install Oracle Database

[oracle@gncasnpl08358 ~]$. db_profile
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[oracle@gncasnpl08358 dbhome_1]$ export DISPLAY=localhost:11.0
[oracle@gncasnpl08358 dbhome_1]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ ./runInstaller -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /u01/app/oraInventory/logs/InstallActions2023-07-25_11-42-20PM/installerPatchActions_2023-07-25_11-42-20PM.log
Launching Oracle Database Setup Wizard...
[FATAL] [INS-35071] Global database name cannot be left blank.
CAUSE: The Global database name was left blank.

ACTION: Specify a value for the Global database name.
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Conforme indicado no Grid Infrastructure, o swap o João Hoffman vai cuidar posteriormente e o NTP não é utilizado na versão 8.0

External Image
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Execução dos scripts:

[root@gncasnpl08358 ~]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@gncasnpl08361 app]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
External Image

ASMCA

Criação dos Diskgroups faltantes com user Grid.

[grid@gncasnpl08358 ~]$. grid_profile
[grid@gncasnpl08358 ~]$ xauth add gncasnpl08358/unix:10 MIT-MAGIC-COOKIE-1 62fdb29b8a741ad74f6536662ad2b3d0
[grid@gncasnpl08358 ~]$ export DISPLAY=localhost:10.0
[grid@gncasnpl08358 ~]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 ~]$ asmca
External Image
Expanda Disk Groups, clique em "Create…", crie o diskgroup AUX no padrão (DATA_AUX_PRM84PRD) e escolha os 4 discos referentes a este Disk Group. Lembre-se de marcar o "External".

External Image
O tamanho dos discos estão trocados, note, que os DATA_AUX, com maior tamanho, deveriam ser os menores, escolhi, de forma correta, pelo tamanho:

External Image
Repita o processo para o ARCH (ARCH_PRM84PRD).

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Status do grid, já com os diskgroups:

```
[grid@gncasnpl08358 ~]$ crsctl status res -t
--------------------------------------------------------------------------------
Name           Target  State        Server                   State details
--------------------------------------------------------------------------------
Local Resources
--------------------------------------------------------------------------------
ora.LISTENER.lsnr
               ONLINE  ONLINE       gncasnpl08358            STABLE
               ONLINE  ONLINE       gncasnpl08361            STABLE
ora.chad
               ONLINE  ONLINE       gncasnpl08358            STABLE
               ONLINE  ONLINE       gncasnpl08361            STABLE
ora.net1.network
               ONLINE  ONLINE       gncasnpl08358            STABLE
               ONLINE  ONLINE       gncasnpl08361            STABLE
ora.ons
               ONLINE  ONLINE       gncasnpl08358            STABLE
               ONLINE  ONLINE       gncasnpl08361            STABLE
--------------------------------------------------------------------------------
Cluster Resources
--------------------------------------------------------------------------------
ora.ARCH_PRM84PRD.dg(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
      2        ONLINE  ONLINE       gncasnpl08361            STABLE
      3        ONLINE  OFFLINE                               STABLE
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
      2        ONLINE  ONLINE       gncasnpl08361            STABLE
      3        OFFLINE OFFLINE                               STABLE
ora.DATA_AUX_PRM84PRD.dg(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
      2        ONLINE  ONLINE       gncasnpl08361            STABLE
      3        ONLINE  OFFLINE                               STABLE
ora.DATA_PRM84PRD.dg(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
      2        ONLINE  ONLINE       gncasnpl08361            STABLE
      3        OFFLINE OFFLINE                               STABLE
ora.LISTENER_SCAN1.lsnr
      1        ONLINE  ONLINE       gncasnpl08361            STABLE
ora.LISTENER_SCAN2.lsnr
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.LISTENER_SCAN3.lsnr
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.asm(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            Started,STABLE
      2        ONLINE  ONLINE       gncasnpl08361            Started,STABLE
      3        OFFLINE OFFLINE                               STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
      2        ONLINE  ONLINE       gncasnpl08361            STABLE
      3        OFFLINE OFFLINE                               STABLE
ora.cvu
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.gncasnpl08358.vip
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.gncasnpl08361.vip
      1        ONLINE  ONLINE       gncasnpl08361            STABLE
ora.qosmserver
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.scan1.vip
      1        ONLINE  ONLINE       gncasnpl08361            STABLE
ora.scan2.vip
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
ora.scan3.vip
      1        ONLINE  ONLINE       gncasnpl08358            STABLE
--------------------------------------------------------------------------------
[grid@gncasnpl08358 ~]$
```

DBCA

Criação do banco de dados:

```
[oracle@gncasnpl08358 ~]$ . db_profile
[oracle@gncasnpl08358 ~]$   xauth add gncasnpl08358/unix:10  MIT-MAGIC-COOKIE-1  62fdb29b8a741ad74f6536662ad2b3d0
[oracle@gncasnpl08358 ~]$  cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$   export DISPLAY=localhost:10.0
[oracle@gncasnpl08358 dbhome_1]$  export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ dbca
```

- 2

INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

Criado em:26/07/2023
Última atualização em:28/07/2023
por:Alcides Souto

Você está aqui:
- Main
- Getnet
- INSTALL ORACLE RAC 19C IN LINUX – PRD (Máquina física)

< Todos os tópicos
External ImageExternal Image
Projeto do PRM contempla 2 RACs, um RAC Ativo no DataCenter Sul e outro RAC no DataCenter Norte. Entre estes dois RACs haverá um Data Guard.

RAC Sul

- GNCASSPL08359
- GNCASSPL08360

RAC Norte

- GNCASNPL08358
- GNCASNPL08361

Este documento registra a instalação do RAC no Norte.

Entrega dos servidores

Servidores foram entregues pelo João Hoffman, tratam-se de máquinas físicas, sendo assim, revisamos diversos paramêtros de memoria, users e demais requisitos Oracle. Ao desenrolar a investigação, notamos que havia um StanAlone instalado em cada uma das máquinas.

Parâmetros de memória:

```
[root@gncasnpl08361 ~]# cat /etc/sysctl.conf
# System default settings live in /usr/lib/sysctl.d/00-system.conf.
# To override those settings, enter new settings here, or in an /etc/sysctl.d/<name>.conf file
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
# Controls IP packet forwarding
net.ipv4.ip_forward = 0
# Controls source route verification
net.ipv4.conf.default.rp_filter = 1
# Do not accept source routing
net.ipv4.conf.default.accept_source_route = 0
# Controls the System Request debugging functionality of the kernel
kernel.sysrq = 0
# Controls whether core dumps will append the PID to the core filename.
# Useful for debugging multi-threaded applications.
kernel.core_uses_pid = 1
# Controls the use of TCP syncookies
net.ipv4.tcp_syncookies = 1
# Disable netfilter on bridges.
net.bridge.bridge-nf-call-ip6tables = 0
net.bridge.bridge-nf-call-iptables = 0
net.bridge.bridge-nf-call-arptables = 0
# Controls the default maxmimum size of a mesage queue
kernel.msgmnb = 65536
# Controls the maximum size of a message, in bytes
kernel.msgmax = 65536
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmall = 4294967296
kernel.shmmni = 4096
kernel.sem = 250 32000 100 256
net.ipv4.ip_local_port_range = 9000 65500
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048586
kernel.shmmax = 270377232384
```

Todos os pacotes necessários, executar nos dois nós. Documentei apenas 1 deles.

```
[root@gncasnpl08358 ~]# dnf install bc binutils elfutils-libelf elfutils-libelf-devel fontconfig-devel glibc glibc-devel ksh libaio libaio-devel libgcc libnsl librdmacm-devel libstdc++ libstdc++-devel libX11
```

libXau libxcb libXi libXrender libXrender-devel libXtst make net-tools nfs-utils python3 python3-configshell python3-rtslib python3-six smartmontools sysstat targetcli unzip xorg-x11-apps chrony tmux xorg-x11-server-utils glibc.i686 glibc-devel.i686 libstdc++-devel.i686 libaio-devel.i686 libXext.i686 libXtst.i686 zlib-devel.i686 -y
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 54 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 43 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 47 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 26 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 28 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 44 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Package bc-1.07.1-5.el8.x86_64 is already installed.
Package binutils-2.30-119.el8.x86_64 is already installed.
Package elfutils-libelf-0.188-3.el8.x86_64 is already installed.
Package elfutils-libelf-devel-0.188-3.el8.x86_64 is already installed.
Package fontconfig-devel-2.13.1-4.el8.x86_64 is already installed.
Package glibc-2.28-225.el8.x86_64 is already installed.
Package glibc-devel-2.28-225.el8.x86_64 is already installed.
Package ksh-20120801-257.el8.x86_64 is already installed.
Package libaio-0.3.112-1.el8.x86_64 is already installed.
Package libaio-devel-0.3.112-1.el8.x86_64 is already installed.
Package libgcc-8.5.0-18.el8.x86_64 is already installed.
Package libnsl-2.28-225.el8.x86_64 is already installed.
Package rdma-core-devel-44.0-2.el8.1.x86_64 is already installed.
Package libstdc++-8.5.0-18.el8.x86_64 is already installed.
Package libstdc++-devel-8.5.0-18.el8.x86_64 is already installed.
Package libX11-1.6.8-5.el8.x86_64 is already installed.
Package libXau-1.0.9-3.el8.x86_64 is already installed.
Package libxcb-1.13.1-1.el8.x86_64 is already installed.
Package libXi-1.7.10-1.el8.x86_64 is already installed.
Package libXrender-0.9.10-7.el8.x86_64 is already installed.
Package libXrender-devel-0.9.10-7.el8.x86_64 is already installed.
Package libXtst-1.2.3-7.el8.x86_64 is already installed.
Package make-1:4.2.1-11.el8.x86_64 is already installed.
Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Package nfs-utils-1:2.3.3-59.el8.x86_64 is already installed.
Package python36-3.6.8-38.module+el8.5.0+12207+5c5719bc.x86_64 is already installed.
Package python3-six-1.11.0-8.el8.noarch is already installed.
Package smartmontools-1:7.1-1.el8.x86_64 is already installed.
Package sysstat-11.7.3-9.el8.x86_64 is already installed.
Package unzip-6.0-46.el8.x86_64 is already installed.
No match for argument: xorg-x11-apps
Package chrony-4.2-1.el8.x86_64 is already installed.
Package xorg-x11-server-utils-7.7-27.el8.x86_64 is already installed.
Error: Unable to find a match: xorg-x11-apps
[root@gncasnpl08358 ~]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 51 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 48 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 43 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 42 kB/s | 2.1 kB 00:00
Zabbix 6.0 37 kB/s | 2.1 kB 00:00
Package xorg-x11-xauth-1:1.0.9-12.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@gncasnpl08358 ~]# yum install xterm
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs) 43 kB/s | 2.8 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - High Availability (RPMs) 47 kB/s | 2.4 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs) 50 kB/s | 2.4 kB 00:00
Zabbix 5.0 frontend 42 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 I386 44 kB/s | 2.1 kB 00:00
Zabbix - Rhel6 x86_64 41 kB/s | 2.1 kB 00:00
Zabbix - Rhel7 44 kB/s | 2.1 kB 00:00
Red Hat Satellite Tools 6.9 for RHEL 8 x86_64 (RPMs) 43 kB/s | 2.1 kB 00:00
Zabbix 6.0 44 kB/s | 2.1 kB 00:00
Dependencies resolved.
==
Package Architecture Version Repository Size
==
Installing:
xterm x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 529 k
Installing dependencies:
libXaw x86_64 1.0.13-10.el8 rhel-8-for-x86_64-appstream-rpms 194 k
xterm-resize x86_64 331-1.el8_3.2 rhel-8-for-x86_64-appstream-rpms 38 k
Installing weak dependencies:
xorg-x11-fonts-misc noarch 7.5-19.el8 rhel-8-for-x86_64-appstream-rpms 5.8 M
Transaction Summary
==
Install 4 Packages
Total download size: 6.5 M
Installed size: 8.8 M
Is this ok [y/N]: y
Downloading Packages:

```
(1/4): xterm-331-1.el8_3.2.x86_64.rpm 5.3 MB/s | 529 kB 00:00
(2/4): xterm-resize-331-1.el8_3.2.x86_64.rpm 1.2 MB/s | 38 kB 00:00
(3/4): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm 26 MB/s | 5.8 MB 00:00
(4/4): libXaw-1.0.13-10.el8.x86_64.rpm 32 kB/s | 194 kB 00:06
----------------------------------------------------------------------------------------------------
Total 1.1 MB/s | 6.5 MB 00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : xterm-resize-331-1.el8_3.2.x86_64 1/4
Installing : libXaw-1.0.13-10.el8.x86_64 2/4
Installing : xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch 3/4
Installing : xterm-331-1.el8_3.2.x86_64 4/4
Running scriptlet: xterm-331-1.el8_3.2.x86_64 4/4
Verifying : xorg-x11-fonts-misc-7.5-19.el8.noarch 1/4
Verifying : libXaw-1.0.13-10.el8.x86_64 2/4
Verifying : xterm-331-1.el8_3.2.x86_64 3/4
Verifying : xterm-resize-331-1.el8_3.2.x86_64 4/4
Installed products updated.
Installed:
libXaw-1.0.13-10.el8.x86_64 xorg-x11-fonts-misc-7.5-19.el8.noarch xterm-331-1.el8_3.2.x86_64 xterm-resize-331-1.el8_3.2.x86_64
Complete!
[root@gncasnpl08358 ~]#
```

Users oracle e grid já estavam configurados também, nos dois nos:

```
[root@gncasnpl08358 ~]# id oracle
uid=54340(oracle) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]# id grid
uid=54341(grid) gid=54324(oinstall) groups=54324(oinstall),54325(dba),54326(oper),54327(asmdba),54328(asmadmin)
[root@gncasnpl08358 ~]#
```

Disable firewall, nos dois nos:

```
[root@gncasnpl08358 ~]# systemctl stop firewalld
[root@gncasnpl08358 ~]#
[root@gncasnpl08358 ~]# systemctl disable firewalld
Removed /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
```

Criação de profiles. Lembre que o nó 1 é PRMPRD1 e ASM1 e nó 2 PRMPRD2 e ASM2:

```
vi /home/oracle/db_profile
export ORACLE_BASE=/u01/app/oracle_19c
export ORACLE_HOME=/u01/app/oracle/product/19.19.0/dbhome_1
export ORACLE_SID=PRMCTG1
export OH=$ORACLE_HOME
export JAVA_HOME=/usr/bin/java
export GGS_HOME=/ggs
export PATH=.:/usr/bin/java/bin:/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/oracle/bin:$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib:$ORACLE_HOME/network/jlib
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$ORACLE_HOME/oracm/lib:/lib:/usr/lib:/usr/local/lib
export LIBPATH=/ggs:$ORACLE_HOME/lib:/usr/lib:$LIBPATH
chmod 775 /home/oracle/db_profile
chown oracle:oinstall /home/oracle/db_profile
vi /home/grid/grid_profile
export ORACLE_BASE=/u01/app/grid_19c
export ORACLE_HOME=/u01/app/19.19.0/grid
export ORACLE_SID=+ASM1;
export OH=$ORACLE_HOME
export PATH=$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
chmod 775 /home/grid/grid_profile
chown grid:oinstall /home/grid/grid_profile
```

Download e unzip dos pacotes, pode baixar nos 2 nós, mas os unzip da home do grid e oracle, somente no nó 01:

```
su - grid
. /home/grid/grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35037840_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35050341_190000_Linux-x86-64.zip
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35004974_190000_Linux-x86-64.zip
```

oracleasm

João já entregou as máquinas com tudo instalado, sendo assim, vamos ter que desinstalar tudo e seguir exatamente como fizemos nos RACs de HML.

```
[root@gncasnpl08358 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
```

```
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
[root@gncasnpl08361 ~]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```

Shutdown dos serviços e deinstall

```
[grid@gncasnpl08358 app]$ . /home/grid/grid_profile
[grid@gncasnpl08358 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08358'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08358' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08358' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08358'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08358' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08358' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[grid@gncasnpl08361 app]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasnpl08361'
CRS-2673: Attempting to stop 'ora.asm' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasnpl08361' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.evmd' on 'gncasnpl08361' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasnpl08361'
CRS-2677: Stop of 'ora.cssd' on 'gncasnpl08361' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasnpl08361' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[root@gncasnpl08358 ~]# . /home/grid/grid_profile
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
```

```
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08358 ~]#
[root@gncasnpl08361 ~]# . /home/grid/grid_profile
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD01
Disk "ARCH_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk ARCH_PRM84PRD02
Disk "ARCH_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD01
Disk "DATA_AUX_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_AUX_PRM84PRD02
Disk "DATA_AUX_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD01
Disk "DATA_PRM84PRD01" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD02
Disk "DATA_PRM84PRD02" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD03
Disk "DATA_PRM84PRD03" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD04
Disk "DATA_PRM84PRD04" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD05
Disk "DATA_PRM84PRD05" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD06
Disk "DATA_PRM84PRD06" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD07
Disk "DATA_PRM84PRD07" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD08
Disk "DATA_PRM84PRD08" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD09
Disk "DATA_PRM84PRD09" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD10
Disk "DATA_PRM84PRD10" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD11
Disk "DATA_PRM84PRD11" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD12
Disk "DATA_PRM84PRD12" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]# oracleasm deletedisk DATA_PRM84PRD13
Disk "DATA_PRM84PRD13" defines an unmarked device
Dropping disk: done
[root@gncasnpl08361 ~]#
[root@gncasnpl08358 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 ~]# oracleasm listdisks
[root@gncasnpl08361 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08361 ~]# oracleasm listdisks
```

Remoção dos binários:

```
[root@gncasnpl08358 ~]# cd /tmp
[root@gncasnpl08358 tmp]# rm -rf *
[root@gncasnpl08358 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08358 tmp]# cd /etc/oracle
[root@gncasnpl08358 oracle]# rm -rf *
[root@gncasnpl08358 oracle]# rm -rf /etc/oratab
[root@gncasnpl08358 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
```

```
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08358 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08358 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08358 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08358 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08358 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08358 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08358 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08358 oracle]# cd /u01/
[root@gncasnpl08358 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 09:20 .
dr-xr-xr-x. 22 root root 4096 Jul 18 16:47 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:40 app
drwx------ 2 root root 16384 Jun 16 17:49 lost+found
[root@gncasnpl08358 u01]# rm -rf app
[root@gncasnpl08358 u01]#
[root@gncasnpl08361 ~]# cd /tmp
[root@gncasnpl08361 tmp]# rm -rf *
[root@gncasnpl08361 tmp]# rm -rf /etc/oraInst.loc
[root@gncasnpl08361 tmp]# cd /etc/oracle
[root@gncasnpl08361 oracle]# rm -rf *
[root@gncasnpl08361 oracle]# rm -rf /etc/oratab
[root@gncasnpl08361 oracle]# rm -rf /tmp/.oracle
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgacfsctl -l ofsctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/ucfgadvmctl -l advmctl
-bash: /usr/lib/methods/ucfgadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefacfsctl -l ofsctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/lib/methods/udefadvmctl -l advmctl
-bash: /usr/lib/methods/udefadvmctl: No such file or directory
[root@gncasnpl08361 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@gncasnpl08361 oracle]# rmrole oracle_devmgmt
bash: rmrole: command not found...
[root@gncasnpl08361 oracle]# setkst
bash: setkst: command not found...
[root@gncasnpl08361 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@gncasnpl08361 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@gncasnpl08361 oracle]# rm -rf /sbin/helpers/acfs
[root@gncasnpl08361 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@gncasnpl08361 oracle]# cd /u01/
[root@gncasnpl08361 u01]# ls -la
total 28
drwxr-xr-x 4 oracle oinstall 4096 Jun 19 14:45 .
dr-xr-xr-x. 22 root root 4096 Jun 16 17:38 ..
drwxrwx--- 8 oracle oinstall 4096 Jul 20 11:41 app
drwx------ 2 root root 16384 Jun 16 17:50 lost+found
[root@gncasnpl08361 u01]# rm -rf app
[root@gncasnpl08361 u01]#
```

Criação dos novos diretórios:

```
mkdir -p /u01/app/oracle/product/19.19.0/dbhome_1
mkdir -p /u01/app/19.19.0/grid
chown -R grid:oinstall /u01/app/19.19.0/
chown -R oracle:oinstall /u01/app/oracle/product/19.19.0
mkdir -p /u01/app/grid_19c
mkdir -p /u01/app/oracle_19c
chown -R grid:oinstall /u01/app/grid_19c/
chown -R oracle:oinstall /u01/app/oracle_19c/
chmod 775 /u01/app/oracle_19c/
chmod 775 /u01/app/grid_19c/
```

Formatação dos discos (que foram apagados previamente do ASM):

```
[root@gncasnpl08358 u01]# fdisk -l
Disk /dev/sda: 446.6 GiB, 479559942144 bytes, 936640512 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disklabel type: gpt
Disk identifier: EFD0B073-FC44-4EF7-B31F-0E3BA6ACFD10
Device Start End Sectors Size Type
/dev/sda1 2048 411647 409600 200M EFI System
/dev/sda2 411648 4605951 4194304 2G Microsoft basic data
/dev/sda3 4605952 6703103 2097152 1G Linux filesystem
/dev/sda4 6703104 936638463 929935360 443.4G Linux LVM
Disk /dev/sdb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdd: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdd1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sde: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sde1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdf: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdf1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdg: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdh: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdh1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdi: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdi1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
```

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdm: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdm1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdn: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdn1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdo: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdo1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdp: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdp1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sds: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sds1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdt: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdt1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdw: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdw1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdx: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdx1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdy: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdy1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdz: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdz1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaa: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdaa1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdac: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdac1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdad: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdad1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdae: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdae1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdaf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdah1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdai: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdai1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdaj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdak: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdak1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdal: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdal1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdam: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdam1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdan: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdan1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdao: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdao1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdap: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdap1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdaq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdaq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdar: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdar1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdas: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdas1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdat: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdat1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdau: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdau1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdav: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdav1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdaw: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdax: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdax1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sday: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sday1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdaz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdaz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdba: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdba1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdbd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbe: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbe1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdbf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdbg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbh: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdbh1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbi: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdbi1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbj: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdbj1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbk: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdbk1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbl: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdbl1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbm: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/sdbm1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbn: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/sdbn1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbo: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/sdbo1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/sdbp: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/sdbp1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbq: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/sdbq1 2048 629145599 629143552 300G 83 Linux
Disk /dev/sdbr: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/sdbs: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/sdbs1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbt: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/sdbt1 2048 83888639 83886592 40G 83 Linux
Disk /dev/sdbu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/sdbu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/sdbv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/sdbw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/sdbx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdby: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/sdby1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdbz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/sdbz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdca: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/sdca1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcb: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/sdcb1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcc: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/sdcc1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcd: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/sdcd1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdce: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/sdce1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcf: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/sdcf1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/sdcg: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/sdcg1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/rhel-root: 70 GiB, 75161927680 bytes, 146800640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-swap: 4 GiB, 4294967296 bytes, 8388608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/rhel-home: 369.4 GiB, 396667912192 bytes, 774742016 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 262144 bytes / 262144 bytes
Disk /dev/mapper/mpatha: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa6cd4bd1
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatha1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathb: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7fdbaa0a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathb1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathq: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8bed29bd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathq1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathr: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d9e03db
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathr1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/vgu01-u01: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/vgu02-u02: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/mpaths: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ac3d4de
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpaths1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpatht: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x87a35585
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpatht1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathu: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfcada6bc
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathu1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathv: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc002c96
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathv1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathw: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x051618b3
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathw1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathx: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa8943581
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathx1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathy: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xbf8b3df6
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathy1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathz: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd37dca50
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathz1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathc: 144 GiB, 154620395520 bytes, 301992960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xdfe3cc8c
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathc1 2048 301992959 301990912 144G 8e Linux LVM
Disk /dev/mapper/mpathaa: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcfb47f11
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaa1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathab: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf767a22a
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathab1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/vg_ggs-ggs: 144 GiB, 154614628352 bytes, 301981696 sectors
Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathac: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd153a1cd
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathac1 2048 629145599 629143552 300G 83 Linux
Disk /dev/mapper/mpathad: 300 GiB, 322122547200 bytes, 629145600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk /dev/mapper/mpathae: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4726b26
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathae1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathaf: 40 GiB, 42950983680 bytes, 83888640 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf4b14040
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathaf1 2048 83888639 83886592 40G 83 Linux
Disk /dev/mapper/mpathag: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6b2b1dac
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathag1 2048 2147485439 2147483392 1024G 83 Linux
Disk /dev/mapper/mpathah: 1 TiB, 1099512545280 bytes, 2147485440 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6f549e4d
Device Boot Start End Sectors Size Id Type
/dev/mapper/mpathah1 2048 2147485439 2147483392 1024G 83 Linux
[root@gncasnpl08358 u01]#
Output do multipath

[root@gncasnpl08358 u01]# multipath -ll
mpatha (360000970000597600012533030304431) dm-3 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:0 sdb 8:16 active ready running
|- 11:0:1:0 sdw 65:96 active ready running
|- 13:0:0:0 sdar 66:176 active ready running
`- 13:0:1:0 sdbm 68:0 active ready running
mpathaa (360000970000597600012533030313542) dm-20 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:20 sdv 65:80 active ready running
|- 11:0:1:20 sdaq 66:160 active ready running
|- 13:0:0:20 sdbl 67:240 active ready running
`- 13:0:1:20 sdcg 69:64 active ready running
mpathab (360000970000597600012533030303435) dm-22 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:3 sde 8:64 active ready running
|- 11:0:1:3 sdz 65:144 active ready running
|- 13:0:0:3 sdau 66:224 active ready running
`- 13:0:1:3 sdbp 68:48 active ready running
mpathac (360000970000597600012533030303436) dm-24 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:4 sdf 8:80 active ready running
|- 11:0:1:4 sdaa 65:160 active ready running
|- 13:0:0:4 sdav 66:240 active ready running
`- 13:0:1:4 sdbq 68:64 active ready running
mpathad (360000970000597600012533030313443) dm-25 EMC,SYMMETRIX
size=300G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:5 sdg 8:96 active ready running
|- 11:0:1:5 sdab 65:176 active ready running
|- 13:0:0:5 sdaw 67:0 active ready running
`- 13:0:1:5 sdbr 68:80 active ready running
mpathae (360000970000597600012533030313444) dm-26 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:6 sdh 8:112 active ready running
|- 11:0:1:6 sdac 65:192 active ready running

```
|- 13:0:0:6 sdax 67:16 active ready running
`- 13:0:1:6 sdbs 68:96 active ready running
mpathaf (360000970000597600012533030313445) dm-27 EMC,SYMMETRIX
size=40G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:7 sdi 8:128 active ready running
|- 11:0:1:7 sdad 65:208 active ready running
|- 13:0:0:7 sday 67:32 active ready running
`- 13:0:1:7 sdbt 68:112 active ready running
mpathag (360000970000597600012533030313446) dm-28 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:8 sdj 8:144 active ready running
|- 11:0:1:8 sdae 65:224 active ready running
|- 13:0:0:8 sdaz 67:48 active ready running
`- 13:0:1:8 sdbu 68:128 active ready running
mpathah (360000970000597600012533030313530) dm-29 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:9 sdk 8:160 active ready running
|- 11:0:1:9 sdaf 65:240 active ready running
|- 13:0:0:9 sdba 67:64 active ready running
`- 13:0:1:9 sdbv 68:144 active ready running
mpathb (360000970000597600012533030304432) dm-4 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:1 sdc 8:32 active ready running
|- 11:0:1:1 sdx 65:112 active ready running
|- 13:0:0:1 sdas 66:192 active ready running
`- 13:0:1:1 sdbn 68:16 active ready running
mpathc (360000970000597600012533030304433) dm-19 EMC,SYMMETRIX
size=144G features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:2 sdd 8:48 active ready running
|- 11:0:1:2 sdy 65:128 active ready running
|- 13:0:0:2 sdat 66:208 active ready running
`- 13:0:1:2 sdbo 68:32 active ready running
mpathq (360000970000597600012533030313531) dm-6 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:10 sdl 8:176 active ready running
|- 11:0:1:10 sdag 66:0 active ready running
|- 13:0:0:10 sdbb 67:80 active ready running
`- 13:0:1:10 sdbw 68:160 active ready running
mpathr (360000970000597600012533030313532) dm-8 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:11 sdm 8:192 active ready running
|- 11:0:1:11 sdah 66:16 active ready running
|- 13:0:0:11 sdbc 67:96 active ready running
`- 13:0:1:11 sdbx 68:176 active ready running
mpaths (360000970000597600012533030313533) dm-11 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:12 sdn 8:208 active ready running
|- 11:0:1:12 sdai 66:32 active ready running
|- 13:0:0:12 sdbd 67:112 active ready running
`- 13:0:1:12 sdby 68:192 active ready running
mpatht (360000970000597600012533030313534) dm-12 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:13 sdo 8:224 active ready running
|- 11:0:1:13 sdaj 66:48 active ready running
|- 13:0:0:13 sdbe 67:128 active ready running
`- 13:0:1:13 sdbz 68:208 active ready running
mpathu (360000970000597600012533030313535) dm-13 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:14 sdp 8:240 active ready running
|- 11:0:1:14 sdak 66:64 active ready running
|- 13:0:0:14 sdbf 67:144 active ready running
`- 13:0:1:14 sdca 68:224 active ready running
mpathv (360000970000597600012533030313536) dm-14 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:15 sdq 65:0 active ready running
|- 11:0:1:15 sdal 66:80 active ready running
|- 13:0:0:15 sdbg 67:160 active ready running
`- 13:0:1:15 sdcb 68:240 active ready running
mpathw (360000970000597600012533030313537) dm-15 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:16 sdr 65:16 active ready running
|- 11:0:1:16 sdam 66:96 active ready running
|- 13:0:0:16 sdbh 67:176 active ready running
`- 13:0:1:16 sdcc 69:0 active ready running
mpathx (360000970000597600012533030313538) dm-16 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:17 sds 65:32 active ready running
```

```
|- 11:0:1:17 sdan 66:112 active ready running
|- 13:0:0:17 sdbi 67:192 active ready running
`- 13:0:1:17 sdcd 69:16 active ready running
mpathy (360000970000597600012533030313539) dm-17 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:18 sdt 65:48 active ready running
|- 11:0:1:18 sdao 66:128 active ready running
|- 13:0:0:18 sdbj 67:208 active ready running
`- 13:0:1:18 sdce 69:32 active ready running
mpathz (360000970000597600012533030313541) dm-18 EMC,SYMMETRIX
size=1.0T features='1 queue_if_no_path' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=1 status=active
|- 11:0:0:19 sdu 65:64 active ready running
|- 11:0:1:19 sdap 66:144 active ready running
|- 13:0:0:19 sdbk 67:224 active ready running
`- 13:0:1:19 sdcf 69:48 active ready running
[root@gncasnpl08358 u01]#
```

Apagamos apenas os discos que foram utilizados na instalação standalone e reapresentamos ao ASM:

```
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathae1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00409259 s, 12.5 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaf1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00649853 s, 7.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathab1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00404717 s, 12.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathac1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00418631 s, 12.2 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathad1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.000105445 s, 486 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathq1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00391171 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathr1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00395635 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpaths1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00358853 s, 14.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpatht1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00451208 s, 11.3 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathu1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00439451 s, 11.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathv1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00373145 s, 13.7 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathw1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00607942 s, 8.4 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathx1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00389783 s, 13.1 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathy1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00399422 s, 12.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathz1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00397336 s, 12.9 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathaa1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00370638 s, 13.8 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathag1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00364786 s, 14.0 MB/s
[root@gncasnpl08358 u01]# dd if=/dev/zero of=/dev/mapper/mpathah1 bs=512 count=100
100+0 records in
```

```
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.00387046 s, 13.2 MB/s
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD01 /dev/mapper/mpathae1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk ARCH_PRM84PRD02 /dev/mapper/mpathaf1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD01 /dev/mapper/mpathab1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_AUX_PRM84PRD02 /dev/mapper/mpathac1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD01 /dev/mapper/mpathq1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD02 /dev/mapper/mpathr1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD03 /dev/mapper/mpaths1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD04 /dev/mapper/mpatht1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD05 /dev/mapper/mpathu1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD06 /dev/mapper/mpathv1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD07 /dev/mapper/mpathw1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD08 /dev/mapper/mpathx1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD09 /dev/mapper/mpathy1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD10 /dev/mapper/mpathz1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD11 /dev/mapper/mpathaa1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD12 /dev/mapper/mpathag1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]# oracleasm createdisk DATA_PRM84PRD13 /dev/mapper/mpathah1
Writing disk header: done
Instantiating disk: done
[root@gncasnpl08358 u01]#
[root@gncasnpl08358 u01]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@gncasnpl08358 u01]# oracleasm listdisks
ARCH_PRM84PRD01
ARCH_PRM84PRD02
DATA_AUX_PRM84PRD01
DATA_AUX_PRM84PRD02
DATA_PRM84PRD01
DATA_PRM84PRD02
DATA_PRM84PRD03
DATA_PRM84PRD04
DATA_PRM84PRD05
DATA_PRM84PRD06
DATA_PRM84PRD07
DATA_PRM84PRD08
DATA_PRM84PRD09
DATA_PRM84PRD10
DATA_PRM84PRD11
DATA_PRM84PRD12
DATA_PRM84PRD13
```

Agora temos discos de ASM zerados e apresentados ao nó 01.
Podemos seguir com as questões de interfaces de rede. Consolidação dos IPs:

External Image
External Image
Resumo da configuração dos IPs:

GNCASNPL08358
IP Público: 180.104.147.12
IP Branca: 22.248.70.14
IP Backup: 22.240.70.12
Interconnect: 172.27.147.11
IP VIP: 180.104.147.15

Scan1: 180.104.147.19
Scan2: 180.104.147.20
Scan3: 180.104.147.21
GNCASNPL08361
IP Público: 180.104.147.14
IP Branca: 22.248.70.16
IP Backup: 22.240.70.15
Interconnect: 172.27.147.12
IP VIP: 180.104.147.18
Resolução das chaves de SSH:
NOTA: o documento de RAC de HML está completo a parte do SSH.

```
passwd oracle
passwd grid
*** usar uma senha padrão
./sshUserSetup.sh -user grid -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
./sshUserSetup.sh -user oracle -hosts "gncasnpl08358 gncasnpl08361" -noPromptPassphrase
chmod 600 /home/grid/.ssh/config
chmod 600 /home/oracle/.ssh/config
```

Configuração do /etc/hosts:

```
### Install RAC Oracle
180.104.147.12 GNCASNPL08358 GNCASNPL08358.getnet.local
22.240.70.12 GNCASNPL08358.bkp GNCASNPL08358.bkp.getnet.local
22.248.70.14 GNCASNPL08358.wt GNCASNPL08358.wt.getnet.local
180.104.147.15 GNCASNPL08358-vip GNCASNPL08358-vip.getnet.local
180.104.147.14 GNCASNPL08361 GNCASNPL08361.getnet.local
22.240.70.15 GNCASNPL08361.bkp GNCASNPL08361.bkp.getnet.local
22.248.70.16 GNCASNPL08361.wt GNCASNPL08361.wt.getnet.local
180.104.147.18 GNCASNPL08361-vip GNCASNPL08361-vip.getnet.local
```

Download do patch que corrige o SSH:

```
su - grid
. grid_profile
cd /u02/oracle/patches
wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_Linux-x86-64.zip
unzip p30159782_195000OCWRU_Linux-x86-64.zip
```

Install Grid Infrastructure

```
[grid@gncasnpl08358 ~]$ . grid_profile
[grid@gncasnpl08358 ~]$ cd $ORACLE_HOME
[grid@gncasnpl08358 grid]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[grid@gncasnpl08358 grid]$ export DISPLAY=localhost:11.0
[grid@gncasnpl08358 grid]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 grid]$ ./gridSetup.sh -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /tmp/GridSetupActions2023-07-20_04-46-39PM/installerPatchActions_2023-07-20_04-46-39PM.log
Launching Oracle Grid Infrastructure Setup Wizard...
[FATAL] [INS-40426] Grid installation option has not been specified.
ACTION: Specify the valid installation option.
[grid@gncasnpl08358 grid]$ ./gridSetup.sh
```

External Image
External Image
External Image
Adicionar o segundo nó do cluster:
Clique em “Add…”

External Image
External Image
External Image
Como trata-se de máquinas físicas, há mais interfaces de redes configuradas, como da idrac, por exemplo, lembre-se de utilizar a interface publica para o public e interconnect para ASM & Private:

External Image
NOTA: O erro abaixo está relacionado a interface do IDrac, conversei com o João HOffman, ele me pediu para ignorar:

External Image
External Image
External Image
External Image
NOTA: lembre-se de ajustar o discovery path:

External Image
Escolhemos apenas discos do +DATA:

External Image
External Image
External Image
External Image
External Image
External Image
External Image
External Image
Observações:
Node Connectivity é a respeito do Idrac, o swap eu combinei do João Hoffman ajustar posteriormente dado a emergencia do projeto e os demais são irrelevantes:

External Image
External Image
External Image
Execução de scripts:

```
[root@gncasnpl08358 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08361 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@gncasnpl08358 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/rootcrs_gncasnpl08358_2023-07-25_05-58-37PM.log
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 17:58:46 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 17:58:46 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 17:58:47 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 17:58:48 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 17:58:56 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 17:58:59 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 17:59:07 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 17:59:10 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 17:59:14 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 17:59:14 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/07/25 17:59:34 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 17:59:38 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 17:59:42 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 17:59:45 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
ASM has been created and started successfully.
[DBT-30001] Disk groups created successfully. Check /u01/app/grid_19c/cfgtoollogs/asmca/asmca-230725PM060015.log for details.
2023/07/25 18:01:21 CLSRSC-482: Running command: '/u01/app/19.19.0/grid/bin/ocrconfig -upgrade grid oinstall'
CRS-4256: Updating the profile
Successful addition of voting disk 25590c0c233d4f26bf21c16dcb0516a4.
Successfully replaced voting disk group with +DATA_PRM84PRD.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
## STATE File Universal Id File Name Disk group
-- ----- ----------------- --------- ---------
1. ONLINE 25590c0c233d4f26bf21c16dcb0516a4 (/dev/oracleasm/disks/DATA_PRM84PRD01) [DATA_PRM84PRD]
Located 1 voting disk(s).
2023/07/25 18:02:34 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:03:40 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:03:40 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:04:42 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:05:00 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@gncasnpl08358 ~]#
[root@gncasnpl08361 ~]# /u01/app/19.19.0/grid/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.19.0/grid
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.19.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/rootcrs_gncasnpl08361_2023-07-25_06-22-36PM.log
2023/07/25 18:22:42 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/07/25 18:22:43 CLSRSC-363: User ignored prerequisites during installation
2023/07/25 18:22:43 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/07/25 18:22:43 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
```

2023/07/25 18:22:44 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/07/25 18:22:44 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/07/25 18:22:45 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/07/25 18:22:46 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/07/25 18:22:47 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/07/25 18:22:48 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/07/25 18:22:49 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/07/25 18:22:50 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/07/25 18:22:51 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
2023/07/25 18:23:23 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/07/25 18:24:05 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/07/25 18:24:05 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/07/25 18:24:35 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/07/25 18:24:51 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded

Install Oracle Database

[oracle@gncasnpl08358 ~]$. db_profile
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$ xauth add gncasnpl08358/unix:11 MIT-MAGIC-COOKIE-1 61ffaebd4b2496351672b043d28ab8fc
[oracle@gncasnpl08358 dbhome_1]$ export DISPLAY=localhost:11.0
[oracle@gncasnpl08358 dbhome_1]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ ./runInstaller -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /u01/app/oraInventory/logs/InstallActions2023-07-25_11-42-20PM/installerPatchActions_2023-07-25_11-42-20PM.log
Launching Oracle Database Setup Wizard...
[FATAL] [INS-35071] Global database name cannot be left blank.
CAUSE: The Global database name was left blank.
ACTION: Specify a value for the Global database name.
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Conforme indicado no Grid Infrastructure, o swap o João Hoffman vai cuidar posteriormente e o NTP não é utilizado na versão 8.0

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Execução dos scripts:

[root@gncasnpl08358 ~]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@gncasnpl08361 app]# /u01/app/oracle/product/19.19.0/dbhome_1/root.sh
Performing root user operation.
The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.19.0/dbhome_1
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
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ASMCA

Criação dos Diskgroups faltantes com user Grid.

[grid@gncasnpl08358 ~]$. grid_profile
[grid@gncasnpl08358 ~]$ xauth add gncasnpl08358/unix:10 MIT-MAGIC-COOKIE-1 62fdb29b8a741ad74f6536662ad2b3d0
[grid@gncasnpl08358 ~]$ export DISPLAY=localhost:10.0
[grid@gncasnpl08358 ~]$ export CV_ASSUME_DISTID=OEL8.1
[grid@gncasnpl08358 ~]$ asmca
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Expanda Disk Groups, clique em “Create…”, crie o diskgroup AUX no padrão (DATA_AUX_PRM84PRD) e escolha os 4 discos referentes a este Disk Group. Lembre-se de marcar o “External”.

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O tamanho dos discos estão trocados, note, que os DATA_AUX, com maior tamanho, deveriam ser os menores, escolhi, de forma correta, pelo tamanho:

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Repita o processo para o ARCH (ARCH_PRM84PRD).

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Status do grid, já com os diskgroups:

[grid@gncasnpl08358 ~]$ crsctl status res -t

Name Target State Server State details

Local Resources
----
ora.LISTENER.lsnr
ONLINE ONLINE gncasnpl08358 STABLE
ONLINE ONLINE gncasnpl08361 STABLE
ora.chad
ONLINE ONLINE gncasnpl08358 STABLE
ONLINE ONLINE gncasnpl08361 STABLE
ora.net1.network
ONLINE ONLINE gncasnpl08358 STABLE
ONLINE ONLINE gncasnpl08361 STABLE
ora.ons
ONLINE ONLINE gncasnpl08358 STABLE
ONLINE ONLINE gncasnpl08361 STABLE

Cluster Resources
----
ora.ARCH_PRM84PRD.dg(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 STABLE
2 ONLINE ONLINE gncasnpl08361 STABLE
3 ONLINE OFFLINE STABLE
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 STABLE
2 ONLINE ONLINE gncasnpl08361 STABLE
3 OFFLINE OFFLINE STABLE
ora.DATA_AUX_PRM84PRD.dg(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 STABLE
2 ONLINE ONLINE gncasnpl08361 STABLE
3 ONLINE OFFLINE STABLE
ora.DATA_PRM84PRD.dg(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 STABLE
2 ONLINE ONLINE gncasnpl08361 STABLE
3 OFFLINE OFFLINE STABLE
ora.LISTENER_SCAN1.lsnr
1 ONLINE ONLINE gncasnpl08361 STABLE
ora.LISTENER_SCAN2.lsnr
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.LISTENER_SCAN3.lsnr
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.asm(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 Started,STABLE
2 ONLINE ONLINE gncasnpl08361 Started,STABLE
3 OFFLINE OFFLINE STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
1 ONLINE ONLINE gncasnpl08358 STABLE
2 ONLINE ONLINE gncasnpl08361 STABLE
3 OFFLINE OFFLINE STABLE
ora.cvu
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.gncasnpl08358.vip
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.gncasnpl08361.vip
1 ONLINE ONLINE gncasnpl08361 STABLE
ora.qosmserver
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.scan1.vip
1 ONLINE ONLINE gncasnpl08361 STABLE
ora.scan2.vip
1 ONLINE ONLINE gncasnpl08358 STABLE
ora.scan3.vip
1 ONLINE ONLINE gncasnpl08358 STABLE

[grid@gncasnpl08358 ~]$

DBCA

Criação do banco de dados:

[oracle@gncasnpl08358 ~]$. db_profile
[oracle@gncasnpl08358 ~]$ xauth add gncasnpl08358/unix:10 MIT-MAGIC-COOKIE-1 62fdb29b8a741ad74f6536662ad2b3d0
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME
[oracle@gncasnpl08358 dbhome_1]$ export DISPLAY=localhost:10.0
[oracle@gncasnpl08358 dbhome_1]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@gncasnpl08358 dbhome_1]$ dbca
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Na aba Sizing, altere para 5000, na acha de “Character sets”, detalhei mais abaixo:

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PSU 19.18

```
[root@gncasnpl08358 ~]# . /home/grid/grid_profile
[root@gncasnpl08358 ~]# cd $ORACLE_HOME/OPatch
[root@gncasnpl08358 OPatch]# export CV_ASSUME_DISTID=OEL8.1
[root@gncasnpl08358 OPatch]# ./opatchauto apply /u02/oracle/patches/34762026

NOTA: minha sessão caiu, não registrei o output

[root@gncasnpl08361 ~]# . /home/grid/grid_profile
[root@gncasnpl08361 ~]# cd $ORACLE_HOME/OPatch
[root@gncasnpl08361 OPatch]# export CV_ASSUME_DISTID=OEL8.1
[root@gncasnpl08361 OPatch]# ./opatchauto apply /u02/oracle/patches/34762026

OPatchauto session is initiated at Wed Jul 26 01:25:50 2023

System initialization log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchautodb/systemconfig2023-07-26_01-25-54AM.log.

Session log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/opatchauto2023-07-26_01-26-09AM.log
The id for this session is YTDS

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.19.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch applicability verified successfully on home /u01/app/19.19.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Executing patch validation checks on home /u01/app/19.19.0/grid
Patch validation checks successfully completed on home /u01/app/19.19.0/grid


Executing patch validation checks on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.19.0/dbhome_1


Verifying SQL patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
SQL patch applicability verified successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Preparing to bring down database service on home /u01/app/oracle/product/19.19.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.19.0/dbhome_1 to bring down database service


Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.19.0/grid
Prepatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/crs_prepatch_apply_inplace_gncasnpl08361_2023-07-26_01-27-21AM.log
CRS service brought down successfully on home /u01/app/19.19.0/grid


Performing prepatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Start applying binary patch on home /u01/app/oracle/product/19.19.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Performing postpatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1
```

```
Start applying binary patch on home /u01/app/19.19.0/grid
Binary patch applied successfully on home /u01/app/19.19.0/grid


Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.19.0/grid
Postpatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/crs_postpatch_apply_inplace_gncasnpl08361_2023-07-2
6_01-39-46AM.log
CRS service started successfully on home /u01/app/19.19.0/grid


Preparing home /u01/app/oracle/product/19.19.0/dbhome_1 after database service restarted
No step execution required.........


Trying to apply SQL patch on home /u01/app/oracle/product/19.19.0/dbhome_1
SQL patch applied successfully on home /u01/app/oracle/product/19.19.0/dbhome_1

OPatchAuto successful.

--------------------------------Summary--------------------------------

Patching is completed successfully. Please find the summary as follows:

Host:gncasnpl08361
RAC Home:/u01/app/oracle/product/19.19.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SKIPPED:

Patch: /u02/oracle/patches/34762026/34768569
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/34762026/33575402
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/34762026/34863894
Reason: This patch is not applicable to this specified target type - "rac_database"


==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/34762026/34765931
Log: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-31-02AM_1.log

Patch: /u02/oracle/patches/34762026/34768559
Log: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-31-02AM_1.log


Host:gncasnpl08361
CRS Home:/u01/app/19.19.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/34762026/33575402
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-35-36AM_1.log

Patch: /u02/oracle/patches/34762026/34765931
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-35-36AM_1.log

Patch: /u02/oracle/patches/34762026/34768559
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-35-36AM_1.log

Patch: /u02/oracle/patches/34762026/34768569
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-35-36AM_1.log

Patch: /u02/oracle/patches/34762026/34863894
```

```
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_01-35-36AM_1.log



OPatchauto session completed at Wed Jul 26 01:55:23 2023
Time taken to complete the session 29 minutes, 29 seconds
[root@gncasnpl08361 OPatch]#
```

JDK 19.18

```
[root@gncasnpl08358 OPatch]# ./opatchauto apply /u02/oracle/patches/34777391

OPatchauto session is initiated at Wed Jul 26 01:56:59 2023

System initialization log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchautodb/systemconfig2023-07-26_01-57-01AM.log.

Session log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/opatchauto2023-07-26_01-57-17AM.log
The id for this session is 1MBP

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.19.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch applicability verified successfully on home /u01/app/19.19.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Executing patch validation checks on home /u01/app/19.19.0/grid
Patch validation checks successfully completed on home /u01/app/19.19.0/grid


Executing patch validation checks on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.19.0/dbhome_1


Verifying SQL patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
No sqlpatch prereq operations are required on the local node for this home
No step execution required.........


Preparing to bring down database service on home /u01/app/oracle/product/19.19.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.19.0/dbhome_1 to bring down database service


Bringing down database service on home /u01/app/oracle/product/19.19.0/dbhome_1
No step execution required.........


Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.19.0/grid
Prepatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/crs_prepatch_apply_inplace_gncasnpl08358_2023-07-26_01-57-45AM.log
CRS service brought down successfully on home /u01/app/19.19.0/grid


Performing prepatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Start applying binary patch on home /u01/app/oracle/product/19.19.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Performing postpatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Start applying binary patch on home /u01/app/19.19.0/grid
Binary patch applied successfully on home /u01/app/19.19.0/grid
```

```
Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.19.0/grid
Postpatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08358/crsconfig/crs_postpatch_apply_inplace_gncasnpl08358_2023-07-2
6_02-02-26AM.log
CRS service started successfully on home /u01/app/19.19.0/grid


Starting database service on home /u01/app/oracle/product/19.19.0/dbhome_1
No step execution required.........


Preparing home /u01/app/oracle/product/19.19.0/dbhome_1 after database service restarted
No step execution required.........


Trying to apply SQL patch on home /u01/app/oracle/product/19.19.0/dbhome_1
No SQL patch operations are required on local node for this home

OPatchAuto successful.

--------------------------------Summary--------------------------------

Patching is completed successfully. Please find the summary as follows:

Host:gncasnpl08358
RAC Home:/u01/app/oracle/product/19.19.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/34777391
Log: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_02-00-55AM_1.log


Host:gncasnpl08358
CRS Home:/u01/app/19.19.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/34777391
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_02-01-42AM_1.log



OPatchauto session completed at Wed Jul 26 02:07:53 2023
Time taken to complete the session 10 minutes, 54 seconds
[root@gncasnpl08358 OPatch]#


[root@gncasnpl08361 OPatch]# ./opatchauto apply /u02/oracle/patches/34777391

OPatchauto session is initiated at Wed Jul 26 02:09:36 2023

System initialization log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchautodb/systemconfig2023-07-26_02-09-37AM.log.

Session log file is /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/opatchauto2023-07-26_02-09-55AM.log
The id for this session is FI2A

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.19.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch applicability verified successfully on home /u01/app/19.19.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Executing patch validation checks on home /u01/app/19.19.0/grid
Patch validation checks successfully completed on home /u01/app/19.19.0/grid
```

```
Executing patch validation checks on home /u01/app/oracle/product/19.19.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.19.0/dbhome_1


Verifying SQL patch applicability on home /u01/app/oracle/product/19.19.0/dbhome_1
No sqlpatch prereq operations are required on the local node for this home
No step execution required.........


Preparing to bring down database service on home /u01/app/oracle/product/19.19.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.19.0/dbhome_1 to bring down database service


Bringing down database service on home /u01/app/oracle/product/19.19.0/dbhome_1
No step execution required.........


Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.19.0/grid
Prepatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/crs_prepatch_apply_inplace_gncasnpl08361_2023-07-26_
02-10-27AM.log
CRS service brought down successfully on home /u01/app/19.19.0/grid


Performing prepatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Start applying binary patch on home /u01/app/oracle/product/19.19.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Performing postpatch operation on home /u01/app/oracle/product/19.19.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.19.0/dbhome_1


Start applying binary patch on home /u01/app/19.19.0/grid
Binary patch applied successfully on home /u01/app/19.19.0/grid


Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.19.0/grid
Postpatch operation log file location: /u01/app/grid_19c/crsdata/gncasnpl08361/crsconfig/crs_postpatch_apply_inplace_gncasnpl08361_2023-07-2
6_02-15-29AM.log
CRS service started successfully on home /u01/app/19.19.0/grid


Starting database service on home /u01/app/oracle/product/19.19.0/dbhome_1
No step execution required.........


Preparing home /u01/app/oracle/product/19.19.0/dbhome_1 after database service restarted
No step execution required.........


Trying to apply SQL patch on home /u01/app/oracle/product/19.19.0/dbhome_1
No SQL patch operations are required on local node for this home

OPatchAuto successful.

--------------------------------Summary--------------------------------

Patching is completed successfully. Please find the summary as follows:

Host:gncasnpl08361
RAC Home:/u01/app/oracle/product/19.19.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:
```

```
Patch: /u02/oracle/patches/34777391
Log: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_02-13-55AM_1.log


Host:gncasnpl08361
CRS Home:/u01/app/19.19.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/34777391
Log: /u01/app/19.19.0/grid/cfgtoollogs/opatchauto/core/opatch/opatch2023-07-26_02-14-44AM_1.log



OPatchauto session completed at Wed Jul 26 02:21:40 2023
Time taken to complete the session 12 minutes, 4 seconds
[root@gncasnpl08361 OPatch]#
```

OJVM 19.18

```
[oracle@gncasnpl08361 ~]$ . db_profile
[oracle@gncasnpl08361 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jul 26 02:22:49 2023
Version 19.18.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.18.0.0.0

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>

[oracle@gncasnpl08361 ~]$ cd $ORACLE_HOME/OPatch
[oracle@gncasnpl08361 OPatch]$ ./opatch apply /u02/oracle/patches/34786990
Oracle Interim Patch Installer version 12.2.0.1.37
Copyright (c) 2023, Oracle Corporation.  All rights reserved.


Oracle Home       : /u01/app/oracle/product/19.19.0/dbhome_1
Central Inventory : /u01/app/oraInventory
   from           : /u01/app/oracle/product/19.19.0/dbhome_1/oraInst.loc
OPatch version    : 12.2.0.1.37
OUI version       : 12.2.0.7.0
Log file location : /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatch/opatch2023-07-26_02-25-11AM_1.log

Verifying environment and performing prerequisite checks...
OPatch continues with these patches:   34786990

Do you want to proceed? [y|n]
y
User Responded with: Y
All checks passed.

Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.
(Oracle Home = '/u01/app/oracle/product/19.19.0/dbhome_1')


Is the local system ready for patching? [y|n]
y
User Responded with: Y
```

```
User Responded with: Y
Backing up files...
Applying interim patch '34786990' to OH '/u01/app/oracle/product/19.19.0/dbhome_1'

Patching component oracle.javavm.server, 19.0.0.0.0...

Patching component oracle.javavm.server.core, 19.0.0.0.0...

Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...

Patching component oracle.rdbms, 19.0.0.0.0...

Patching component oracle.javavm.client, 19.0.0.0.0...
Patch 34786990 successfully applied.
Log file location: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatch/opatch2023-07-26_02-25-11AM_1.log

OPatch succeeded.
[oracle@gncasnpl08361 OPatch]$ srvctl config database
PRMCTG
[oracle@gncasnpl08361 OPatch]$ srvctl start database -d PRMCTG

[oracle@gncasnpl08358 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jul 26 02:24:17 2023
Version 19.18.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.18.0.0.0

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.18.0.0.0
[oracle@gncasnpl08358 ~]$ cd $ORACLE_HOME/OPatch
[oracle@gncasnpl08358 OPatch]$ ./opatch apply /u02/oracle/patches/34786990
Oracle Interim Patch Installer version 12.2.0.1.37
Copyright (c) 2023, Oracle Corporation.  All rights reserved.


Oracle Home       : /u01/app/oracle/product/19.19.0/dbhome_1
Central Inventory : /u01/app/oraInventory
   from           : /u01/app/oracle/product/19.19.0/dbhome_1/oraInst.loc
OPatch version    : 12.2.0.1.37
OUI version       : 12.2.0.7.0
Log file location : /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatch/opatch2023-07-26_02-31-25AM_1.log

Verifying environment and performing prerequisite checks...
OPatch continues with these patches:   34786990

Do you want to proceed? [y|n]
y
User Responded with: Y
All checks passed.

Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.
(Oracle Home = '/u01/app/oracle/product/19.19.0/dbhome_1')


Is the local system ready for patching? [y|n]
y
User Responded with: Y
Backing up files...
Applying interim patch '34786990' to OH '/u01/app/oracle/product/19.19.0/dbhome_1'
```

```
Patching component oracle.javavm.server, 19.0.0.0.0...

Patching component oracle.javavm.server.core, 19.0.0.0.0...

Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...

Patching component oracle.rdbms, 19.0.0.0.0...

Patching component oracle.javavm.client, 19.0.0.0.0...
Patch 34786990 successfully applied.
Log file location: /u01/app/oracle/product/19.19.0/dbhome_1/cfgtoollogs/opatch/opatch2023-07-26_02-31-25AM_1.log

OPatch succeeded.
[oracle@gncasnpl08358 OPatch]$
[oracle@gncasnpl08358 OPatch]$ srvctl start database -d PRMCTG
```

Add Service

```
[oracle@gncasnpl08361 admin]$ srvctl add service -d PRMCTG -s PRMNRTCTGSRV -r PRMCTG1,PRMCTG2
[oracle@gncasnpl08361 admin]$ srvctl start service -d PRMCTG -s PRMNRTCTGSRV
[oracle@gncasnpl08361 admin]$ srvctl status  service -d PRMCTG -s PRMNRTCTGSRV
Service PRMNRTCTGSRV is running on instance(s) PRMCTG1,PRMCTG2
```

Archivelog

```
[oracle@gncasnpl08358 ~]$ srvctl stop database -d PRMCTG
[oracle@gncasnpl08358 ~]$ srvctl start database -d PRMCTG -o mount

[oracle@gncasnpl08358 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jul 26 02:56:47 2023
Version 19.18.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.18.0.0.0

SQL> alter database archivelog;

Database altered.

[oracle@gncasnpl08358 ~]$  srvctl stop database -d PRMCTG
[oracle@gncasnpl08358 ~]$ srvctl start database -d PRMCTG

[oracle@gncasnpl08358 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jul 26 03:02:31 2023
Version 19.18.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.18.0.0.0

SQL> alter system set log_archive_dest_1='LOCATION=+ARCH_PRM84PRD' scope=both sid='*';

System altered.

SQL> archive log list
Database log mode              Archive Mode
Automatic archival             Enabled
Archive destination            +ARCH_PRM84PRD
Oldest online log sequence     27
Next log sequence to archive   28
Current log sequence           28
SQL>
```